

A Justification for Equal Sharing

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Abstract

We study the evolution of productive traits and surplus-sharing norms in matching markets with heterogeneous agents. Partnerships form endogenously, output depends on partners' productivity, and social norms govern surplus division. We compare market sharing, equal sharing, and unequal sharing. Although equal sharing can generate inefficient matches in the short run, it promotes productive traits and achieves long-run efficiency comparable to market sharing. Unequal sharing, by contrast, can distort both matching and evolutionary selection, allowing inefficient traits and outcomes to persist. We also study the evolution of sharing norms over time, providing an evolutionary rationale for the persistence of equal sharing.

Keywords: Equal sharing, norms, matching, evolutionary game theory

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1 Introduction

The equal sharing norm advocates for the equitable distribution of resources among individuals. It has deep roots in human history and continues to influence contemporary societal structures. For example, the ancient principle of equal sharing in partnership law, as documented in the *Institutes of Justinian* since 533 CE, remains widely adopted by modern law firms (Smith, 2001). Anthropological evidence suggests that egalitarian sharing norms were also central to many hunter–gatherer societies, where food obtained through hunting or gathering was typically distributed broadly within the camp rather than retained by the individual who produced it (Lee, 1979; Woodburn, 1982). Consistent with these observations, experimental evidence from fifteen small-scale societies shows that individuals often display strong fairness and sharing preferences rather than purely self-interested behavior (Henrich et al., 2001). Equal sharing of royalties among band members is also a common practice, even among the most renowned bands (Polcz, 2023). Finally, the kibbutz has been a vibrant modern Israeli community that promotes equality in resource allocation among community members (Abramitzky, 2008, 2011, 2018).

While equal sharing offers appealing benefits, such as fostering fairness and reinforcing group cohesion, it is inherently rigid. Market-based approaches, which allow for compensation to vary based on different factors, are often seen as more efficient and adaptable to changing conditions. Despite this, equal sharing remains a common practice in certain areas of the market economy. This raises the question: what underlying mechanisms sustain the persistence of equal sharing, even when more flexible alternatives are available? In this paper, we present an evolutionary explanation.

We model a heterogeneous population of agents who endogenously form partnerships. The output of each partnership depends on the productivity of the matched agents. The division of this output is governed by societal sharing norms.

We analyze three types of sharing norms. First, the *market sharing norm* allows agents to split output based on their market opportunities, modeled as a perfectly transferable utility (TU) one-sided matching market. Second, the *equal sharing norm* requires matched pairs to divide the output equally. Third, we consider *unequal sharing norms*, for example, sharing rules that allocate a larger (or smaller) share to the higher ranked agent in the social hierarchy. Unlike the market sharing norm, equal and unequal sharing norms lack flexibility, causing the market to operate as a nontransferable utility (NTU) one-sided matching market.

In our model, evolution operates at three levels: agents’ matching behavior, the distribution of productive traits in the population, and the sharing norms. At the behavioral level, agents within a generation participate in a matching process that immediately results in a stable matching. At the productivity level, the distribution of productive traits evolves more slowly across generations over

time according to the replicator dynamic, which is driven by the payoffs earned through matching and converges to a stable distribution. Finally, at the societal level, sharing norms evolve at the slowest rate, shaped by competition among multiple communities with different sharing norms based on their societal performances.

We find that, although the equal sharing norm may yield stable matchings that are less efficient than those under the market sharing norm at the behavioral level, both norms still lead to efficiency at the level of trait evolution. The reason is that the market sharing rule attains efficiency through flexibility (by compensating individuals appropriately), whereas equal sharing attains it through robustness (by matching individuals who jointly produce the most to the greatest extent possible). As a result, both ultimately favor the selection of the distribution of productive traits that generates the highest partnership outputs.

In contrast, unequal sharing norms can introduce inefficiencies in the process of trait evolution, as traits conducive to productivity may be suppressed when they are at a disadvantage within a hierarchical society. We show that unequal sharing norms result in production inefficiency in the matching stage, and less productive traits survive the evolutionary process. As a result, over time, unequal sharing norms are largely outperformed by both the market sharing norm and the equal sharing norm, which will prevail in the long run through the evolution of sharing norms.

The baseline model assumes that match surplus depends only on agents' abilities. This assumption is useful because it isolates the effect of sharing norms on surplus division and the evolution of productive traits. In many real partnerships, however, surplus is not generated automatically. Agents must exert costly effort, and effort choices may not be perfectly contractible. Hence, the way surplus is divided may affect whether the potential surplus from a match is realized in the first place. To capture this possibility, we extend the model to allow for moral hazard. This extension shows that equal sharing and market sharing can differ once incentive problems are introduced. Equal sharing can sustain efficiency when production is not distorted by moral hazard, but it may fail when equal division does not preserve effort incentives. Market sharing, by contrast, can eliminate the moral hazard problem through transfers. Thus, moral hazard provides a more realistic setting in which the equal sharing norm need not be as enduring as the market sharing norm.

To the best of our knowledge, this paper is the first to study how different surplus sharing norms shape long-run evolutionary outcomes in a matching market with agents of heterogeneous abilities. Our paper connects three strands of literature. First, it relates to the theory of evolutionary dynamics in population games. A large literature studies how strategy distributions evolve under payoff-monotonic dynamics, such as the replicator dynamic, where strategies with above-average payoffs increase in frequency over time. [Sandholm \(2010\)](#) provides a unified treatment of these models and their stability properties in large populations. Second, our work is related to the evolutionary study of social norms governing surplus division. [Skyrms \(2014\)](#) analyzes the evolution

of division rules in bargaining environments with random matching. In contrast, our framework features heterogeneous agents whose partnerships form endogenously through a matching process, which fundamentally shapes the evolutionary selection of traits. Finally, our model connects evolutionary dynamics with the theory of matching markets. Classic results in transferable-utility matching, beginning with [Becker \(1973\)](#) and the assignment game of [Shapley and Shubik \(1972\)](#), characterize stable and efficient allocations in matching environments. We embed this matching structure into an evolutionary framework and show how alternative sharing norms interact with matching behavior to determine long-run population outcomes.

Previous studies adopt a similar approach to studying the evolution of cultural or preference traits under rapid matching and a fixed sharing rule (e.g., [Wu and Zhang, 2021](#); [Hiller, Wu, and Zhang, 2023](#); [Wang and Wu, 2025](#)). However, they do not investigate how different sharing rules shape long-run outcomes differently. Our noncooperative setup (moral hazard) is related to the literature on the evolution of sharing norms in the Nash demand game and in repeated games ([Young, 1993](#); [Kranton, 1996](#); [Juang and Sabourian, 2021](#); [Jindani, 2022](#)).

In our model, partnership formation occurs in a one-sided pairwise matching market (i.e., the roommate market) with a continuum of agents in every period. Although many standard results from the parallel two-sided matching settings—e.g., the equivalence of socially efficient and stable matchings, the convexity of stable payoffs ([Shapley and Shubik, 1972](#); [Gretsky, Ostroy, and Zame, 1992](#))—extend to our one-sided setting, there is no paper that explicitly states and proves these results.¹ To this end, we provide results in more detail and prove them rigorously, which may serve as a reference for subsequent papers that model the evolution of behavior, productivity, culture, and/or preferences in one-sided matching markets.

The only other work we are aware of that provides evolutionary justifications for the equal sharing norm is a book by [Skyrms \(2014\)](#). In his model, a population of agents is randomly matched to play a dollar division game. The population consists of three types of agents: super-greedy agents who demand at least $2/3$, fair-minded agents who demand at least $1/2$, and super-modest agents who demand at least $1/3$. The author demonstrates that, under the replicator dynamic, a homogeneous population of fair-minded agents and a polymorphic population with equal proportions of super-greedy and super-modest agents are both asymptotically stable. However, the basin of attraction for the fair-minded population is significantly larger than that of the polymorphic population. The intuition is straightforward. Super-greedy agents secure $2/3$ only when paired with super-modest agents and receive nothing otherwise. Super-modest agents can always reach an agreement with any opponent, but they consistently receive the smallest share. Fair-minded

¹A few papers—[Azevedo, Weyl, and White \(2013\)](#); [Chiappori, Salanié, and Galichon \(2019\)](#); [Carmona and Lao-hakunakorn \(2024\)](#)—provide existence results in similar roommate matching settings but do not discuss properties such as the convexity and compactness of the set of stable payoffs, which are crucial to our evolutionary existence results.

agents, by contrast, typically secure 1/2, especially when there is a sufficient number of fair-minded or super-modest agents in the population. Our model differs from [Skyrms \(2014\)](#) in several key aspects. First, agents in our model exhibit heterogeneity in their productivity rather than in demand rules. Second, partnerships in our model emerge endogenously, as opposed to being formed through random encounters, and this is key to the long-run efficiency of the equal sharing norm in our evolutionary setting.

The rest of the paper is organized as follows. Section 2 studies the market sharing norm. Section 3 studies alternative fixed-proportion sharing norms, i.e., equal and unequal sharing norms. Section 4 considers the long-run evolutionary stability of different sharing norms. Section 5 extends the model to incorporate moral hazard. Section 6 concludes. The appendix collects omitted proofs.

2 The Market Sharing Norm

2.1 Population and Matching

Consider a unit mass of agents. Each agent is characterized by a productive trait $i \in N = \{1, 2, \dots, n\}$. Let $x_i \in [0, 1]$ denote the mass of trait- i agents, for any $i \in N$ and $\sum_{i=1}^n x_i = 1$. Let $x = (x_1, x_2, \dots, x_n)$ denote the population state. In this paper, we use $\mathbb{R}_+$ to refer to the set of nonnegative real numbers and $\mathbb{R}_{++}$ to refer to the set of strictly positive real numbers. Let $X = \{x \in \mathbb{R}_+^n \mid \sum_{i=1}^n x_i = 1\}$ denote the set of population states; it is an $(n - 1)$ -simplex Δ^{n-1} . Let $X_{++} = \text{int}(X) = \{x \in \mathbb{R}_{++}^n \mid \sum_{i=1}^n x_i = 1\}$ denote the *interior* of X , i.e., the set of population states such that there is a positive mass of each trait.

Agents can find partners to form pairs; they can also choose not to. If an agent remains unmatched, she receives a payoff of 0.² If a trait- i agent and a trait- j agent are matched, they jointly produce a strictly positive payoff of $f(i, j)$. Let $f : N^2 \rightarrow \mathbb{R}_{++}$ denote the production function. Assume that f is symmetric: $f(i, j) = f(j, i)$ for any $i, j \in N$.

Let $\mu_{ij}(x) \geq 0$ denote the mass of (i, j) pairs given the population state x . We have $\mu_{ij}(x) = \mu_{ji}(x)$. Since each (i, i) pair contains two trait- i agents, and each (i, j) pair with $i \neq j$ contains one trait- i agent, we have $2\mu_{ii}(x) + \sum_{j \neq i} \mu_{ij}(x) \leq x_i$. Let $\mu(x)$ be an $n \times n$ matrix that denotes the aggregate matching ([Echenique et al., 2013](#)), whose (i, j) entry is $\mu_{ij}(x)$. Note that the notion of aggregate matching is defined on the population level as opposed to the more familiar individual level. We adopt the stability concept in two-sided TU (transferable utility) matching market ([Shapley and Shubik, 1972](#); [Becker, 1973](#); [Gretsky, Ostroy, and Zame, 1992](#)) to our setting

²The analysis can allow for trait-specific positive outside options, as long as each trait's outside option is below the payoff it can obtain from its worst feasible match.

of one-sided matching market with a continuum of agents to characterize the stable outcome of the partnership formation process.

Definition 1 A *stable outcome* is described by a *stable matching* $\mu^*(x)$ and a vector of *stable payoffs* $w^*(x) = (w_1^*(x), w_2^*(x), \dots, w_n^*(x)) \in \mathbb{R}^n$, such that

- (i) (**individual rationality**) each person gets at least as much as being unmatched: $w_i^*(x) \geq 0$ for any $i \in N$;
- (ii) (**surplus efficiency**) each pair exactly divides the surplus: $w_i^*(x) + w_j^*(x) = f(i, j)$ if $\mu_{ij}^*(x) > 0$, for any $i, j \in N$; and
- (iii) (**no blocking pair**) no pair can receive a sum that is strictly more than the sum of their current payoffs: $w_i^*(x) + w_j^*(x) \geq f(i, j)$ for any $i, j \in N$.

Let us note some implications of the definition. First, since an agent gets a payoff of zero if she remains unmatched and would produce a strictly positive surplus with any other agent, all agents are matched in any stable matching. Second, in the definition, all agents with the same trait have the same stable payoff. This is not an assumption, but would be implied by the definition, namely by the no-blocking-pair condition: If two agents with the same trait have different payoffs, then there is room for a blocking pair to emerge.

Lemma 1 A stable outcome exists.

We show in the proof of Lemma 1 that, just as in a two-sided matching market (Gretsky, Ostroy, and Zame, 1992), a matching is stable if and only if it is socially efficient. Hence, we can characterize a stable matching by the solution to the maximization problem of the total payoff:

$$\max_{\mu(x)} \sum_{i \in N} \sum_{j \leq i} \mu_{ij}(x) f(i, j) \text{ such that } 2\mu_{ii}(x) + \sum_{j \neq i} \mu_{ij}(x) \leq x_i \text{ for all } i \in N.$$

When the solution to the maximization problem is unique, it must be the unique stable matching. We will assume that the stable matching is always unique in the subsequent analysis; Assumption 1(ii) below provides a sufficient condition on the primitives that achieves this.

We make the following additional assumptions on the primitives of the model to eliminate knife-edge cases and simplify the analysis.

Assumption 1

- (i) $f(i, i) \neq f(j, j)$ for any $i \neq j$.

(ii) $f(i, j) + f(i', j') \neq f(i, i') + f(j, j')$ for any $i, i', j, j' \in N$ such that $\{i, i'\} \cap \{j, j'\} = \emptyset$.

Assumption 1(i) ensures that some of the key relevant comparisons are strict. Assumption 1(ii) characterizes a sufficient condition to ensure a unique stable matching.³

Let $W^*(x)$ be the collection of stable payoff vectors $w^*(x)$ that support the unique stable matching $\mu^*(x)$.

To illustrate the stable outcome with an example, we consider the setting with two traits.

Example 1 (Stable outcome of the two-trait economy under the market sharing norm)

Consider the setting with two traits, $N = \{1, 2\}$.

First, suppose $2f(1, 2) < f(1, 1) + f(2, 2)$. The unique stable outcome involves positive assortative matching (PAM)— $\mu_{ii}^* = x_i/2$ for $i \in N$ —and an equal split— $w_i^*(x_i) = f(i, i)/2$ for $i \in N$, as the agents in each pair are of the same type and must receive the same payoff in the stable outcome.

Next, suppose instead $2f(1, 2) > f(1, 1) + f(2, 2)$. The stable outcome involves negative assortative matching (NAM). When $x_1 < 1/2$, the unique stable matching is given by $\mu_{12}^*(x) = x_1$ and $\mu_{22}^*(x) = (x_2 - x_1)/2$, as, given the excess type-2 agents, each type-1 agent matches with a type-2 agent, and the excess type-2 agents pair with each other. The unique stable payoff vector is given by $w^*(x) = (f(1, 2) - f(2, 2)/2, f(2, 2)/2)$; type-2 agents' stable payoffs are pinned down as a positive mass of them form pairs and split their output $f(2, 2)$ to get $f(2, 2)/2$ and type-1 agents who pair with type-2 agents receive the rest of their output $f(1, 2)$. When $x_1 > 1/2$, i.e., there are more type-1 agents than type-2 agents, the unique stable matching is given by $\mu_{12}^*(x) = x_2$ and $\mu_{11}^*(x) = (x_1 - x_2)/2$, and the unique stable payoff vector is given by $w^*(x) = (f(1, 1)/2, f(1, 2) - f(1, 1)/2)$. When $x_1 = 1/2$, the unique stable matching $\mu^*(x)$ is given by $\mu_{12}^*(x) = 1/2$, as each pair involves a type-1 agent and a type-2 agent, and any payoff vector $w^*(x)$ such that $w_1^*(x) + w_2^*(x) = f(1, 2)$ and $f(i, i)/2 \leq w_i^*(x) \leq f(1, 2) - f(j, j)/2$ for $i \in \{1, 2\}$ and $j \neq i$ is stable; the inequalities guarantee that agents do not have an incentive to block the pairing with an agent of the same type.⁴

³If we randomly generate $f(i, j)$ for each pair of i and j from distributions without mass, these conditions are satisfied with probability one. Assumption 1(ii) is sufficient but not necessary for the uniqueness of efficient matchings. If $f(i, j) + f(i', j') = f(i, i') + f(j, j')$ holds for some types $i, i', j, j' \in N$, and if all four types exist, then we could have an efficient matching with pairings of (i, j) and (i', j') , or another efficient matching with pairings of (i, i') and (j, j') . Of course, if some types $i, i', j, j' \in N$ do not exist, then $f(i, j) + f(i', j') = f(i, i') + f(j, j')$ does not result in such multiplicity. In addition, if the efficient match does not involve any pairings of (i, j) , (i', j') , (i, i') , and (j, j') , then $f(i, j) + f(i', j') = f(i, i') + f(j, j')$ does not lead to multiplicity of efficient matches.

⁴This case shows that the stable payoffs can be discontinuous at $x_1 = x_2 = 1/2$. When one type is strictly scarcer, the long side receives its same type matching value, while the scarcer type receives the residual surplus. At $x_1 = x_2 = 1/2$, however, stable payoffs are set-valued. This sharp change in stable payoffs reflects the scarcity-rent logic of the frictionless TU matching model, not a literal prediction that small demographic changes cause immediate discontinuous changes in observed surplus division. In applications, bargaining frictions, institutional constraints, repeated relationships, and gradual adjustment would typically smooth the transition.

Finally, if $2f(1, 2) = f(1, 1) + f(2, 2)$, any matching is efficient and can be supported in a stable outcome. Our Assumption 1(ii) eliminates our consideration of this knife-edge case. ■

2.2 The Evolution of Traits

The distribution of traits evolves over time, guided by the payoffs associated with different traits. We assume that the process of forming partnerships reaches a stable outcome more rapidly than the evolution of traits. Hence, when the distribution of traits shifts, the stable outcome swiftly adapts.⁵

We consider the replicator dynamic to describe the evolution of traits. The replicator dynamic can be derived from a biological growth model (Taylor and Jonker, 1978), an intergenerational cultural transmission model (Bisin and Verdier, 2001), or learning models based on imitation (Sandholm, 2010). Let $\bar{w}^*(x) = \sum_{j \in N} x_j w_j^*(x)$ denote the average payoff in a stable outcome. Note that since there is a unit mass of agents, the average payoff equals the total payoff: $\bar{w}^*(x) = \sum_{i \in N} \sum_{j \leq i} \mu_{ij}^*(x) f(i, j)$. Since the unique stable matching $\mu^*(x)$ must maximize the total payoff, $\bar{w}^*(x)$ does not vary across $w^*(x) \in W^*(x)$. Let $\bar{\mathbf{w}}^*(x) = (\bar{w}^*(x), \bar{w}^*(x), \dots, \bar{w}^*(x)) \in \mathbb{R}^n$ be the n -dimensional vector in which each of the n entries equals $\bar{w}^*(x)$. The replicator dynamic is given by a differential inclusion:⁶

$$\dot{x} \in V^{RDI}(x) \equiv \{x \cdot [w^*(x) - \bar{\mathbf{w}}^*(x)] \mid w^*(x) \in W^*(x)\}. \quad (\text{RDI})$$

Hence, the fraction of agents who carry a trait that is associated with an above-average payoff grows and vice versa.

The dynamic described in (RDI) is a differential inclusion because of the possible indeterminacy of stable payoff vectors. When there are multiple stable payoff vectors, the path the dynamic takes depends on the realized stable payoff vector. For example, at x , if the realized stable payoff vector is $w^*(x)$, then the growth rate of trait i is given by $\dot{x}_i = x_i \cdot [w_i^*(x) - \bar{w}^*(x)]$ for any $i \in N$, which takes the standard form of the replicator dynamic.

We have the following theorem from Sandholm (2010) that provides sufficient conditions for the existence of solution trajectory.

Theorem 1 (Sandholm, 2010, Theorem 6.A.1) *Consider a differential inclusion $\dot{x} \in V(x)$. If V is nonempty, bounded, convex-valued, and upper-hemicontinuous, then for any initial condition $\xi \in \mathbb{R}^n$, there exists a **Carathéodory solution trajectory** $\{x^t\}_{t \in [0, T]}$ to the differential inclusion.*

⁵This approach parallels the indirect evolutionary approach that studies preference evolution (Güth and Yaari, 1992; Güth, 1995), where agents' preferences evolve at a much slower pace compared to their behaviors.

⁶A differential inclusion is a generalization of a differential equation where the derivative of a function is constrained to belong to a set of possible values rather than being determined by a single value.

That is, the solution trajectory is Lipschitz continuous, and its derivative $\dot{x}_t \in V(x)$ at almost all times $t \in [0, T]$.

The following lemma shows that the replicator dynamic we consider is well-behaved and, thus, combined with Theorem 1, has a solution trajectory in the interior of X . Note that the uniqueness of the solution trajectory is not guaranteed.

Lemma 2 *The correspondence V^{RDI} is nonempty-valued, bounded, convex-valued, and upper-hemicontinuous in X_{++} .*

2.3 Analysis

We now study the convergence and stability of the dynamic processes. Convergence informs us where the dynamic process ends up from all initial conditions, and stability determines whether the convergent states of the population are robust against perturbations.⁷

Definition 2 *The ξ -limit set $\omega(\xi)$ is the set of all population states such that the solution trajectory $\{x^t\}_{t \geq 0}$ starting from $x^0 = \xi$ approaches arbitrarily closely infinitely often. Formally,*

$$\omega(\xi) := \left\{ y \in X : \exists \{t_k\}_{k=1}^{\infty} \text{ with } \lim_{k \rightarrow \infty} t_k = \infty \text{ s.t. } x^0 = \xi \text{ and } \lim_{k \rightarrow \infty} x^{t_k} = y \right\}.$$

An element in a ξ -limit set is called a ξ -limit point.

The collection of all limit points is denoted by the set $\Omega = \cup_{\xi \in X} \omega(\xi)$: It captures the notion of recurrence of the dynamics. Let $A \subseteq X$ be a closed set, and call $O \subseteq X$ a neighborhood of A if it is open relative to X and contains A .

Definition 3 *Set $A \subset X$, a proper subset of X , is **Lyapunov stable** if for every neighborhood O of A , there exists a neighborhood O' of A such that every solution trajectory $\{x^t\}_{t \geq 0}$ that starts in O' is contained in O , that is, $x^0 \in O'$ implies that $x^t \in O$ for all $t \geq 0$. If A is not Lyapunov stable, we call it **Lyapunov unstable**.*

Intuitively, Lyapunov stability requires that any displacement from A does not lead the process to go “far” from A at any point in time.

Definition 4 *Set $A \subset X$ is **attracting** if there is a neighborhood $Y \subseteq X$ of A such that every solution trajectory that starts in Y converges to A , that is, $\xi \in Y$ implies $\omega(\xi) \subseteq A$. Set $A \subset X$ is **globally attracting** if it is attracting with $Y = X$. The set of points $\xi \in X$ such that $\omega(\xi) \subseteq A$ is called the **basin of attraction** of A .*

⁷See Sandholm (2010) for a textbook treatment of these concepts and related results.

Intuitively, that A is attracting requires that given any displacement from A , the process returns to A in the limit.

Definition 5 Set $A \subset X$ is **asymptotically stable** if it is Lyapunov stable and attracting. Set $A \subset X$ is **globally asymptotically stable** if it is Lyapunov stable and globally attracting.

Intuitively, the asymptotic stability of A requires that, given any displacement from A , the process never travels “far” from A and returns to A in the limit. The limiting behavior of deterministic dynamics can be characterized as follows. We will focus on the case in which there exists a unique match that leads to the highest payoff.

Proposition 1 (i) Suppose $\arg\max_{(k,l)} f(k,l) = \{(i,i)\}$ for some $i \in N$. The population state $x^* = e_i$, the i th standard basis vector, is the unique globally asymptotically stable state under the replicator dynamic with respect to $X \setminus \{x | x_i = 0\}$. It is also the efficient state. (ii) Suppose $\arg\max_{(k,l)} f(k,l) = \{(i,j)\}$ for some i and $j \neq i$. The population state x^* such that $x_i^* = x_j^* = 0.5$ is the unique globally asymptotically stable state under the replicator dynamic with respect to $X \setminus \{x | x_i = 0 \text{ or } x_j = 0 \text{ or both}\}$. It is also the efficient state.

Proposition 1 shows that the population states that maximize the average payoff in stable matching are asymptotically stable. See Appendix A for Theorems 2 and 3 that we use to prove stability and convergence based on the Lyapunov method. The key is to find a Lyapunov function. We will define the Lyapunov functions for different cases.

Note that in case (ii), when $x_i^* = x_j^* = 1/2$, stable payoffs are not unique. Nevertheless, since the equal-division payoff belongs to the set of stable payoffs, the dynamic inclusion would converge to and settle at the efficient population.

The following example concretely illustrates how Proposition 1 holds when there are two productive traits. In particular, the example illustrates that supermodularity or submodularity does not determine the efficient evolutionarily stable state.

Example 2 (Globally asymptotically stable two-trait state under the market sharing norm)

Consider the setting with two traits: 1 and 2. Without loss of generality, assume that $f(1,1) > f(2,2)$. Hence, the two traits differ in own-match productivity. When $2f(1,2) > f(1,1) + f(2,2)$, we call the payoff structure **submodular**. There is always a unique stable matching in which the mass of cross-trait matching is maximized (NAM). When $2f(1,2) < f(1,1) + f(2,2)$, we call the payoff structure **supermodular**. There is always a unique stable matching with only own-trait matching (PAM).

Case 1. Suppose $f(1,2) > f(1,1)$. This immediately implies submodularity. In this case, we always have a unique stable payoff as long as $x_1 \neq 1/2$. When $x_1 < 1/2$, $w_1(x) = f(1,2) -$

$f(2,2)/2 > w_2(x) = f(2,2)/2$. When $x_1 > 1/2$, $w_1(x) = f(1,1)/2 < w_2(x) = f(1,2) - f(1,1)/2$. Hence, for (RDI), $\dot{x}_1 > 0$ when $0 < x_1 < 1/2$; $\dot{x}_1 < 0$ when $x_1 > 1/2$, which guarantees the globally asymptotic stability of $x^* = (1/2, 1/2)$ given any interior initial state.

Case 2. Suppose $f(1,1) > f(1,2)$ and $2f(1,2) > f(1,1) + f(2,2)$. We still have submodularity. When $x_1 < 1/2$, $w_1(x) = f(1,2) - f(2,2)/2 > w_2(x) = f(2,2)/2$. When $x_1 > 1/2$, $w_1(x) = f(1,1)/2 > w_2(x) = f(1,2) - f(1,1)/2$. Hence, for (RDI), $\dot{x}_1 > 0$ when $x_1 > 0$, which guarantees the globally asymptotic stability of $x_1 = 1$ in $(0, 1]$.

Case 3. Suppose $f(1,1) > f(1,2)$ and $2f(1,2) < f(1,1) + f(2,2)$. In this case, we have supermodularity. We always have $w_1(x) = f(1,1)/2 > w_2(x) = f(2,2)/2$. Hence, for (RDI), $\dot{x}_1 > 0$ when $x_1 > 0$, which guarantees the globally asymptotic stability of $x_1 = 1$ in $(0, 1]$.

To summarize this two-trait example, while supermodularity guarantees that trait 1, the more productive trait, prevails through evolution, submodularity does not guarantee the coexistence of both traits. Only when cross-trait matching leads to the highest production can both traits coexist. These results confirm Proposition 1, that the population state maximizing the average payoff in stable matching is the long-run prediction of the evolutionary process. When $f(1,2)$ is the highest, $x = (1/2, 1/2)$ maximizes the average payoff. When $f(1,1)$ is the highest, $x = (1, 0)$ maximizes the average payoff. ■

3 Sharing Norms and Efficiency

In this section, rather than letting agents determine surplus division endogenously within the market, we introduce exogenous sharing norms that specify how the surplus is distributed in each match. These norms reflect the cultural conventions that governed resource allocation in society before the emergence of free markets.

Let u_{ij} denote a trait- i agent's payoff from her match with a trait- j agent. $u_{ij} + u_{ji} = f(i, j)$. We adopt the stability concept in two-sided NTU (nontransferable utility) matching market (Gale and Shapley, 1962) to our setting of one-sided matching markets with a continuum of agents to characterize the stable matching of the partnership formation process.

Definition 6 A *stable matching* $\mu^*(x)$ satisfies

- (i) (**individual rationality**) each person gets at least as much as staying unmatched: $u_{ij} \geq 0$ for any $i, j \in N$ such that $\mu_{ij}^*(x) > 0$;
- (ii) (**no blocking pair**) no pair of persons can both be strictly better off than what they have in their current matches: $u_{ij} \leq u_{ik}$ or $u_{ji} \leq u_{jl}$ for any $i, j, k, l \in N$ such that $\mu_{ik}^*(x) > 0$ and $\mu_{jl}^*(x) > 0$.

Carmona and Laohakunakorn (2024) prove the existence of stable matching. For x and $\mu^*(x)$, let $u_i^*(x)$ denote the average payoff of all trait- i agents. Let $u^*(x) = (u_1^*(x), u_2^*(x), \dots, u_n^*(x)) \in \mathbb{R}^n$ denote the vector of average payoffs. Let $\bar{u}^*(x)$ denote the average payoff in the entire population. Let $\bar{\mathbf{u}}^*(x) = (\bar{u}^*(x), \bar{u}^*(x), \dots, \bar{u}^*(x)) \in \mathbb{R}^n$. The evolution of traits is still dictated by the replicator dynamic. But now it is given by a differential equation:

$$\dot{x} = V^{RDE}(x) \equiv \{x \cdot [u^*(x) - \bar{\mathbf{u}}^*(x)]\}. \quad (\text{RDE})$$

Theorem 4 in Appendix A establishes asymptotic stability for (RDE).

3.1 Equal Sharing Norm

First, suppose two agents in a pair equally split the payoff they jointly produce. That is, $u_{ij} = f(i, j)/2$. Without loss of generality, assume $f(1, 1) > f(i, i)$ for any $i \neq 1$.

Proposition 2 (i) Suppose $\arg\max_{(k,l)} f(k, l) = \{(1, 1)\}$. Then $x^* = e_1$, where e_1 is the first standard basis vector, is globally asymptotically stable under (RDE) with respect to $X \setminus \{x | x_1 = 0\}$. It is also the efficient state. (ii) Suppose $\arg\max_{(k,l)} f(k, l) = \{(i, j)\}$ for some i and $j \neq i$. Then x^* such that $x_i^* = x_j^* = 1/2$ is globally asymptotically stable under (RDE) with respect to $X \setminus \{x | x_i = 0 \text{ or } x_j = 0 \text{ or both}\}$. It is also the efficient state.

Although Propositions 1 and 2 apply to different matching environments, they rely on the same underlying evolutionary logic. In both cases, efficiency is achieved because traits that are most productive when matched together earn an average payoff above the population average, which is the key condition for the Lyapunov argument. What differs between the two settings is how this payoff advantage is generated. Under NTU matching, as in the equal sharing norm, the most productive traits match with each other to the greatest extent possible, directly raising their average payoff above the population average. Under TU matching, as in the market sharing norm, these traits need not match with each other to the greatest extent possible. Instead, the no-blocking-pair condition ensures that their average payoff still exceeds the population average.

To illustrate, we consider settings with three traits.

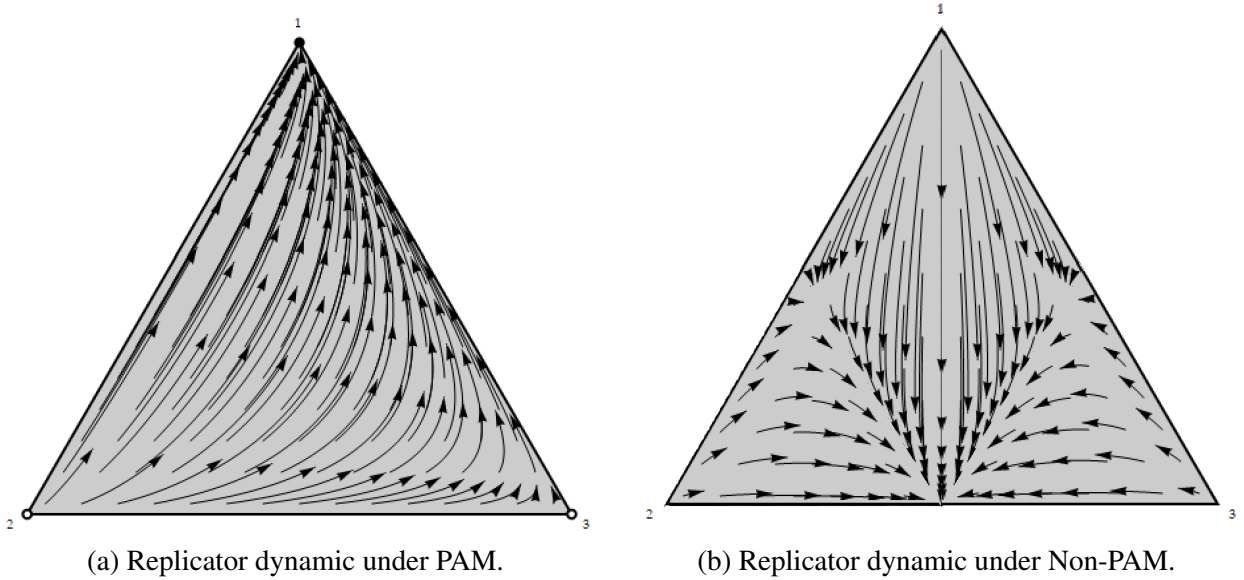
Example 3 (Evolutionarily stable state under the equal sharing norm with three traits)

Consider the setting with three traits: 1, 2, and 3. Suppose that $f(1, 1) > f(1, 2)$, $f(1, 1) > f(1, 3)$, $f(2, 2) > f(1, 2)$, $f(2, 2) < f(2, 3)$, $f(3, 3) < f(1, 3)$, and $f(3, 3) > f(2, 3)$. This payoff structure leads to PAM and the globally attracting and stable state $x^* = (1, 0, 0)$. Figure 1a provides a graphic illustration of (RDE).

It is worth noting that, given the parameter values used to generate Figure 1a, the market sharing norm would not yield PAM. Instead, in the stable matching, all trait-1 agents are matched with each other, and traits 2 and 3 are cross-matched to the greatest extent possible. This matching generates a higher total payoff than PAM under the equal sharing norm, illustrating that equal sharing can be inefficient at the behavioral level. Nevertheless, under equal sharing the payoffs to traits 1, 2, and 3 are 3.25, 1, and 2, respectively. Under market sharing, the corresponding payoffs are 3.25, 1.5, and 2 when $x_2 < x_3$, and 3.25, 1, and 2.5 when $x_2 > x_3$. Hence, regardless of the sharing norm, the replicator dynamic converges to $x^* = (1, 0, 0)$, although the speed of convergence may differ across norms.

Alternatively, suppose $f(1, 1) < f(1, 2) < f(1, 3)$, $f(2, 2) < f(1, 2) < f(2, 3)$, and $f(3, 3) < f(1, 3) < f(2, 3)$. Suppose the initial state $x^0 = (x_1^0, x_2^0, x_3^0) = (0.3, 0.5, 0.2)$. The unique stable matching is characterized by $\mu_{12} = 0.3, \mu_{23} = 0.2$. Then the payoffs under stable matching are $u_1^*(x^0) = f(1, 2)/2$, $u_2^*(x^0) = 0.6 \cdot [f(1, 2)/2] + 0.4 \cdot [f(2, 3)/2]$, and $u_3^*(x^0) = f(2, 3)/2$. The average payoff is $\bar{u}^*(x^0) = 0.6 \cdot [f(1, 2)/2] + 0.4 \cdot [f(2, 3)/2]$. Hence, under the replicator dynamic, $\dot{x}_1 = x_1^0 \cdot [u_1^*(x^0) - \bar{u}^*(x)] < 0$, $\dot{x}_2 = x_2^0 \cdot [u_2^*(x^0) - \bar{u}^*(x)] = 0$, and $\dot{x}_3 = x_3^0 \cdot [u_3^*(x^0) - \bar{u}^*(x)] > 0$. Hence, x_1 shrinks, x_2 stagnates, and x_3 grows. Therefore, $\lim_{t \rightarrow \infty} x^t = (0, 0.5, 0.5)$. Figure 1b provides a graphic illustration of (RDE). ■

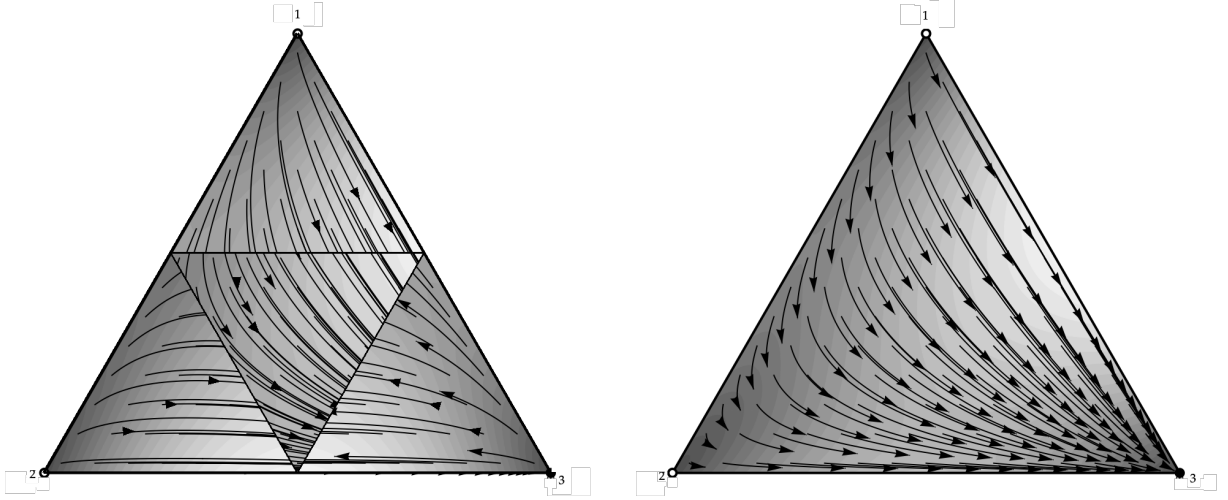
Figure 1: Replicator dynamics under PAM and Non-PAM given the equal sharing norm.



Note: Panel (a) illustrates the replicator dynamic under a setting with PAM: $f(1, 1) = 6.5$, $f(1, 2) = 1$, $f(1, 3) = 5$, $f(2, 2) = 2$, $f(2, 3) = 3.5$, and $f(3, 3) = 4$. The replicator dynamic globally converges to the efficient state $x_1 = 1$. Panel (b) illustrates the replicator dynamics of a setting with non-PAM: $f(1, 1) = 1$, $f(1, 2) = 3$, $f(1, 3) = 5$, $f(2, 2) = 2$, $f(2, 3) = 6$, and $f(3, 3) = 4$. The replicator dynamic globally converges to the efficient state $x_2 = x_3 = 1/2$.

To capture the differences in the evolution of productive traits under different sharing norms, we also depict the replicator dynamic for the numerical example given in Figure 1b under the market sharing norm and an unequal sharing norm in Figure 2.

Figure 2: The replicator dynamic under the market sharing norm and unequal sharing norm.



(a) Replicator dynamic under the market sharing norm (b) Replicator dynamic under an unequal sharing norm

Note: In Panel (a)'s upper region ($x_1 > 1/2$), all trait-2 and trait-3 agents are matched with trait-1 agents, whereas the remaining trait-1 agents are matched with each other, resulting in $w_1(x) = 0.5, w_2(x) = 2.5, w_3(x) = 4.5$. In Panel (a)'s lower left region ($x_2 > 1/2$), all trait-1 and trait-3 agents are matched with trait-2 agents, whereas the remaining trait-2 agents are matched with each other, resulting in $w_1(x) = 2, w_2(x) = 1, w_3(x) = 5$. In Panel (a)'s lower right region ($x_3 > 1/2$), all trait-1 and trait-3 agents are matched with trait-3 agents, whereas the remaining trait-3 agents are matched with each other, resulting in $w_1(x) = 3, w_2(x) = 4, w_3(x) = 2$. In Panel (a)'s middle region (no type is majority), the unique stable outcome is given by $\mu_{12}(x) = (x_1 + x_2 - x_3)/2, \mu_{13} = (x_1 + x_3 - x_2)/2, \mu_{23}(x_2 + x_3 - x_1)/2, w_1(x) = 1, w_2(x) = 2, w_3(x) = 4$. The replicator dynamic globally converges to the efficient state $x_2 = x_3 = 1/2$. Panel (b) illustrates the replicator dynamics given a particular unequal sharing norm, which satisfies $u_{11} = 0.5, u_{12} = 2.9, u_{21} = 0.1, u_{13} = 4.9, u_{31} = 0.1, u_{22} = 1, u_{23} = 5.9, u_{32} = 0.1, u_{33} = 2$. The unique stable matching is PAM. The replicator dynamic globally converges to an inefficient state x where $x_3 = 1$.

3.2 Unequal Sharing Norms

Now suppose the society has a universal sharing norm that features an unequal division of joint production. Let $\alpha_{ij} \in (0, 1)$ denote the share of joint production a trait- i agent receives when matched with a trait- j agent such that $\alpha_{ji} = 1 - \alpha_{ij}$.

Assume that the universally unequal sharing norm takes the following form: $\alpha_{ii} = 0.5$ for any i and $\alpha_{ij} > 0.5$ if $i < j$. The society is hierarchical in this case, where traits also represent hereditary

social ranks and a higher ranked agent can receive a larger share when matched with a lower ranked agent. Assume $f(k, k) = \max_i f(i, i)$, which indicates that trait- k is the most productive in terms of matching with its own type. We allow $k \neq 1$ to maintain generality and avoid assuming that the highest ranked trait is also the most productive in own-trait matching.

For illustration, consider the following two-trait examples, which demonstrate that universally unequal sharing norms can lead to inefficiency through evolution.

Example 4 (Efficient states being unstable under the unequal sharing norm)

Suppose there are two traits: 1 and 2. The payoff structure satisfies that $f(1, 1) = 6$, $f(2, 2) = 8$ and $f(1, 2) = 10$, and the sharing norm satisfies that $\alpha_{12} = 0.9$ and $\alpha_{21} = 0.1$. In this case, $u_{11} = 3$, $u_{22} = 4$, $u_{12} = 9$, $u_{21} = 1$ induces PAM, so the evolutionary dynamic is simply driven by the comparison of u_{11} and u_{22} , which leads to a unique stable state $(0, 1)$. However, this state is not efficient because society can achieve a higher average payoff in the state $(0.5, 0.5)$.

Suppose there are two traits, 1 and 2. The payoff structure satisfies that $f(1, 1) = 6$, $f(2, 2) = 7$ and $f(1, 2) = 10$, and the sharing norm satisfies that $\alpha_{12} = 0.6$ and $\alpha_{21} = 0.4$. In this case, $u_{11} = 3$, $u_{22} = 3.5$, $u_{12} = 6$, $u_{21} = 4$, which induces NAM. To determine the stable state(s), we compare the average payoffs of the trait-1 and trait-2 agents. When $x_1 \leq 0.5$,

$$u_1^*(x) = u_{12} = 6 > u_2^*(x) = \frac{x_1}{x_2}u_{21} + \frac{x_2 - x_1}{x_2}u_{22} = 4\frac{x_1}{x_2} + 3.5\frac{x_2 - x_1}{x_2}.$$

Hence, no state is stable in this region. When $x_1 > 0.5$,

$$u_1^*(x) = \frac{x_1 - x_2}{x_1}u_{11} + \frac{x_2}{x_1}u_{12} = 3\frac{x_1 - x_2}{x_1} + 6\frac{x_2}{x_1}, \quad u_2^*(x) = u_{21} = 4.$$

Hence, there exists a unique stable state $(3/4, 1/4)$. However, this state is not efficient, as society can achieve a higher average payoff in the state $(0.5, 0.5)$. ■

We have the following general result.

Proposition 3 *For any universally unequal sharing norm, there exists a payoff structure f that leads to an asymptotically stable state that is inefficient.*

Next, we ask the question of whether, for a given payoff structure f , there exists a universally unequal sharing norm that leads to an asymptotically stable state that is efficient. We obtain the following negative results.

Proposition 4 *(i) Suppose $\operatorname{argmax}_{(k,l)} f(k, l) = \{(i, j)\}$ for some i and $j \neq i$. Under any unequal sharing norm $\{\alpha_{ij}\}$, the efficient population state x^* with $x_i^* = x_j^* = 0.5$ is not stable. (ii) Suppose*

$\operatorname{argmax}_{(k,l)} f(k,l) = \{(i,i)\}$ for some i . There exists an unequal sharing norm such that the efficient population state $x^* = e_i$ is the unique globally asymptotically stable state, and agents equally share their output.

Proposition 4 shows that (i) when efficiency requires heterogeneity, unequal sharing norms necessarily lead to inefficiency; and (ii) when efficiency requires homogeneity, certain unequal sharing norms can still lead to efficiency. However, in the efficient state, only one trait survives, and all agents sharing this trait equally share their outputs with their partners, so effectively, no unequal sharing norm is observed. In other words, under a fixed unequal sharing norm, the evolutionarily stable outcome is either (i) an inefficient state with an unequal sharing norm or (ii) an efficient state with an equal sharing norm observed.

We use a two-trait example, again, to illustrate Proposition 4. In particular, the efficient state is not necessarily evolutionarily stable.

Example 5 (Evolutionarily stable state under the unequal sharing norm with two traits)

Consider the setting with two traits: 1 and 2.

Case 1. Suppose $f(1,2) > f(1,1)$ and $f(1,2) > f(2,2)$. If $f(1,1)/2 > f(1,2)\alpha_{12}$ or $f(2,2)/2 > f(1,2)\alpha_{21}$ or both, then we have PAM, and (RDE) converges to a homogeneous state, which must be inefficient. If $f(1,2)\alpha_{12} > f(1,1)/2$ and $f(1,2)\alpha_{21} > f(2,2)/2$, then we have NAM. However, at the efficient state x^* such that $x_1^* = x_2^* = 1/2$ and $u_1(x^*) = f(1,2)\alpha_{12} \neq f(1,2)\alpha_{21} = u_2(x^*)$, it implies that x^* is unstable under (RDE).

Case 2. Suppose $f(1,1) > f(1,2)$ or $f(2,2) > f(1,2)$ or both. Also assume that $f(1,1) \neq f(2,2)$. Let $\alpha_{12} = 1/2 + \varepsilon$ and $\alpha_{21} = 1/2 - \varepsilon$, for some $\varepsilon \in (0, 1/2)$. Then there always exists ε such that $f(1,1)/2 > f(1,2)(1/2 + \varepsilon)$ or $f(2,2)/2 > f(1,2)(1/2 - \varepsilon)$ or both. Such ε induces PAM, leading (RDE) to the efficient state e_1 (resp. e_2) if $f(1,1) > f(2,2)$ (resp. $f(2,2) > f(1,1)$). This demonstrates that as long as ε is sufficiently small (i.e., the unequal sharing norm is close to equal sharing), efficiency can be sustained in this case. ■

In summary, the analysis of equal and unequal sharing norms reveals a more general mechanism: any sharing rule under which the evolutionary dynamic leads to the efficient population must satisfy two properties. First, when inefficient types are present, the type(s) in the most productive match must receive above-average payoffs so that the population of inefficient types declines. Second, at the limiting efficient population, the surviving type(s) must receive equal payoffs so that the dynamic can be stationary. Equal sharing guarantees both properties. Unequal sharing can also satisfy them, for example, when the hierarchy embedded in the sharing rule is aligned with the productivity structure. However, when hierarchy is misaligned with productivity, unequal sharing may reward the wrong type and fail to lead the dynamic to the efficient population. Thus, the

advantage of equal sharing is not that it is the only exogenous sharing rule capable of supporting efficiency, but that it does so without requiring social rank to track productive contributions.

4 The Evolution of Sharing Norms

To isolate how different sharing rules shape the evolution of productive types, we have fixed sharing norms in the baseline model. We extend the analysis in this section by allowing the sharing norms themselves to evolve. The extension treats productive types and sharing norms as objects of selection at different levels. Productive types are individual-level traits: individuals carry their productive capacities with them across matches, communities, and generations. Sharing norms, by contrast, are community-level institutions. They are not portable in the same sense; rather, they govern the interactions that take place within the community in which they are embedded. Within each community, productive types evolve according to the baseline replicator dynamics under that community's prevailing sharing norm. Across communities, institutional selection operates through differential growth, survival, and exit. Communities governed by better-performing sharing norms expand or persist, while communities governed by worse-performing norms shrink, dissolve, or are replaced, either because their members move elsewhere or because they leave fewer descendants. This multilevel interpretation follows the logic of [Bowles, Choi, and Hopfensitz \(2003\)](#), who model the coevolution of individual behaviors and group-level institutions in a deme-structured population.⁸

Let s denote a sharing norm, and let $\mathbf{S}$ denote the finite set of available sharing norms. For each $s \in \mathbf{S}$, there is one community governed by norm s , so the number of communities is equal to $|\mathbf{S}|$. Together, these communities constitute the society. Let x^s denote a population state and X^s denote the set of population states in community s . We introduce a scalar function $V_s : X^s \rightarrow \mathbb{R}$, where $V_s(x^s)$ is the average payoff of community s at state x^s .⁹

Let q_s denote the relative size of the community s , defined as its size divided by the total size of the society. The norm-level replicator dynamic is given by

$$\dot{q}_s = \varepsilon q_s [V_s(x^s) - \bar{V}] , \text{ where } \bar{V} = \sum_{r \in \mathbf{S}} q_r V_r(x^r). \quad (\text{RDS})$$

Here ε measures the speed of institutional selection relative to type evolution. Let q_M , q_E , and q_N

⁸A deme-structured population is a population divided into smaller, partially isolated subpopulations (called “demes”). Individuals mate randomly within their own deme, but migration occasionally occurs between groups.

⁹Recall that previously we used $\bar{w}^*(x)$ to denote the average payoff under the market sharing norm. This object is well defined: at each type distribution x , the stable matching is unique, and the average payoff is invariant across stable payoff vectors associated with that matching. Under non-market-sharing norms, we used $\bar{u}^*(x)$ to denote the average payoff. This object is also well defined since the stable matching is unique and the sharing rule pins down the payoffs generated by that matching.

denote the relative sizes of the communities with the market sharing, equal sharing, and unequal sharing norms.

Proposition 5 *Suppose there is an initial full support of sharing norms: $q_s(0) > 0$ for all $s \in \mathbf{S}$. Suppose a unique maximizer: $\{(i, j)\} = \operatorname{argmax}_{(k, l)} f(k, l)$ for some i and $j \neq i$. Then,*

$$\lim_{t \rightarrow \infty} q_M(t) > 0, \lim_{t \rightarrow \infty} q_E(t) > 0, \text{ and } \lim_{t \rightarrow \infty} q_N(t) \geq 0.$$

Proposition 5 shows that equal sharing can coexist with market sharing when norms evolve through institutional selection. The key object is the cumulative performance gap between communities governed by different norms. Market sharing has a transitional advantage because it achieves short-run efficiency at each type distribution, whereas equal sharing reaches efficiency only in the long run. However, since the cumulative transition gap is finite and both norms generate the same limiting community performance, this advantage only changes their relative sizes by a finite amount. It does not eliminate equal sharing. On the other hand, since unequal sharing norms may not achieve efficiency under certain payoff structures (e.g., when $f(i, j) = \max_{(k, l)} f(k, l)$ for some i and $j \neq i$), they can die out via institutional selection.

5 Moral Hazard

So far, we have assumed that when a type- i agent is matched with a type- j agent, the pair generates a joint payoff $f(i, j)$ that is determined solely by their types. In reality, however, partnerships often involve post-match interactions. For instance, matched agents may need to exert effort to produce output. As a result, the final output depends not only on the agents' productive types but also on the effort each agent contributes after the match is formed. Depending on the sharing rule, this creates the possibility of post-match moral hazard: even if high effort by both agents maximizes joint surplus, each agent may have a private incentive to exert lower effort when effort is costly and imperfectly contractible.

To characterize the post-match interactions, we model that each (i, j) pair plays a simultaneous-move game G_{ij} . In this game, each agent can choose an effort level $e \in \mathcal{E}$, where $\mathcal{E}$ is the finite set of feasible pure actions. The pair's joint output is given by $\pi_{ij}(e_i, e_j)$. We assume symmetry: relabeling the agents does not change the joint output, so $\pi_{ij}(e_i, e_j) = \pi_{ji}(e_j, e_i)$. Each effort level is associated with a cost. The cost function is given by $c_i(\cdot)$ for a type- i agent. Let $\Sigma = \Delta\mathcal{E}$ denote the set of mixed strategies. The joint output function $\pi_{ij}(\sigma_i, \sigma_j)$ and the cost function $c_i(\sigma_i)$ extend naturally to mixed strategies by taking expectations over pure strategy profiles.

Fixing a matching $\mu(x)$, for any i, j , let $\gamma_{ij}(x) \in \Delta(\Sigma^2)$ describe the distribution of (mixed) strategy pairs played across matches between type- i and type- j agents. A strategy profile $\gamma(x) =$

$(\gamma_{ij}(x))$ is a vector of distributions of strategy pairs that satisfies the following exchangeability condition: Let $\rho : \Sigma^2 \rightarrow \Sigma^2$ be a mapping that switches the order of strategies, i.e., $\rho(\sigma_i, \sigma_j) = (\sigma_j, \sigma_i)$; then we have $\gamma_{ij}(x)[S] = \gamma_{ji}(x)[\rho(S)]$ for any measurable set $S \subseteq \Sigma^2$.

For illustration purposes, we will first discuss the equal sharing norm and then the market sharing norm. As we will show, the equal sharing norm may give rise to moral hazard, whereas the market sharing norm does not.¹⁰

5.1 The Equal Sharing Norm

Assumption 2 *Two matched agents equally share the joint output that they produce.*

Assumption 2 differs from the equal sharing norm introduced earlier. Under the earlier norm, the two matched agents equally share the joint payoff. Under Assumption 2, by contrast, they equally share the gross joint output while each agent bears their own effort cost. This distinction is crucial for the emergence of moral hazard. Because an agent receives only half of the marginal output generated by their effort but bears the full marginal cost, they may have an incentive to free ride on their partner.

To properly define stable matching, we first need to modify the definition of a blocking pair given post-match interactions. An outcome of the match-to-interact process is characterized by $(\mu(x), \gamma(x))$, a matching profile, and the corresponding strategy profile.

Definition 7 *Fix an outcome $(\mu(x), \gamma(x))$. We say there is a **blocking pair** if there exist types $(i, j) \in N^2$ and a strategy pair $(\sigma_i, \sigma_j) \in \Sigma^2$ such that for some types $k \in N$ and $l \in N$ and strategy pairs $(\sigma'_i, \sigma'_k) \in \Sigma^2$ and $(\sigma'_j, \sigma'_l) \in \Sigma^2$, we have*

- (i) (**positive mass**) $\mu_{ik}(x) > 0$, $\mu_{jl}(x) > 0$, $(\sigma'_i, \sigma'_k) \in \text{supp}\{\gamma_{ik}(x)\}$, and $(\sigma'_j, \sigma'_l) \in \text{supp}\{\gamma_{jl}(x)\}$;
- (ii) (**credibility**) $\sigma_i \in \text{argmax}_{\tilde{\sigma}_i \in \Sigma} [\frac{1}{2}\pi_{ij}(\tilde{\sigma}_i, \sigma_j) - c_i(\tilde{\sigma}_i)]$ and $\sigma_j \in \text{argmax}_{\tilde{\sigma}_j \in \Sigma} [\frac{1}{2}\pi_{ji}(\tilde{\sigma}_j, \sigma_i) - c_j(\tilde{\sigma}_j)]$;
- (iii) (**profitability**) $\frac{1}{2}\pi_{ij}(\sigma_i, \sigma_j) - c_i(\sigma_i) > \frac{1}{2}\pi_{ik}(\sigma'_i, \sigma'_k) - c_i(\sigma'_i)$ and $\frac{1}{2}\pi_{ji}(\sigma_j, \sigma_i) - c_j(\sigma_j) > \frac{1}{2}\pi_{jl}(\sigma'_j, \sigma'_l) - c_j(\sigma'_j)$.

¹⁰We do not separately discuss unequal sharing norms from the equal sharing norm. With moral hazard, a particular unequal sharing rule could, of course, happen to divide the surplus in a way that satisfies the relevant incentive constraints. Unlike market sharing, however, any fixed sharing norm has no mechanism for adjusting the surplus division to those constraints. Its incentive compatibility, therefore, depends on the particular sharing rule and payoff environment rather than following systematically from the norm itself. Since unequal sharing already need not induce efficiency in the baseline model, moral hazard does not generate a new force in its favor.

Condition (i) implies that there is a **positive mass** of agents who participate in a blocking pair. Condition (ii) requires that the deviating agents agree on a Nash equilibrium of G_{ij} so that their strategies are mutual best responses; that is, the deviation is **credible**. Finally, condition (iii) means that the deviating agents strictly prefer to rematch, which indicates that the proposed strategy pair is indeed **profitable**.

Definition 8 A **stable outcome** is described by a **stable matching** $\mu^*(x)$ and a **stable strategy profile** $\gamma^*(x)$ that satisfies the following three conditions.

- (i) (**individual rationality**) Each person gets at least as much as being unmatched: For any $i \in N$ and $j \in N$ such that $\mu_{ij}^*(x) > 0$ and $(\sigma_i, \sigma_j) \in \text{supp}\{\gamma_{ij}^*(x)\}$, $\frac{1}{2}\pi_{ij}(\sigma_i, \sigma_j) - c_i(\sigma_i) \geq 0$;
- (ii) (**internal stability**) For any $i \in N$ and $j \in N$ such that $\mu_{ij}^*(x) > 0$, and for any $(\sigma_i, \sigma_j) \in \text{supp}\{\gamma_{ij}^*(x)\}$, (σ_i, σ_j) is a Nash equilibrium of G_{ij} ; and
- (iii) (**external stability**) There is no blocking pair.

This modified notion of a stable outcome describes an outcome that, once reached, does not induce further strategic or coalitional adjustments.

In what follows, we provide an example to illustrate how moral hazard can arise under the equal sharing norm and how it affects the long-run evolutionary outcome.

Example 6 (Evolutionarily stable state under the equal sharing norm with two traits)

Consider the setting with two traits: 1 and 2. Each agent can choose between two effort levels: H and L . Assume the cost function is identical across different types of agents and $c(H) > c(L)$. The output function $\pi_{ij}(e_i, e_j)$ increases in both arguments.

In addition, we make the following assumptions about the payoff structures. Assume that $\pi_{12}(H, H) - 2c(H)$ is the unique highest joint payoff. This assumption implies that the efficient state is $x_1 = x_2 = 1/2$ with $\mu_{12}(x) = 1/2$.

Assume that $\frac{1}{2}\pi_{12}(L, H) - c(L) > \frac{1}{2}\pi_{12}(H, H) - c(H)$ and $\frac{1}{2}\pi_{12}(L, L) - c(L) > \frac{1}{2}\pi_{12}(L, H) - c(H)$. This implies that if a type-1 agent is matched with a type-2 agent, effort L is their dominant action. This is where the moral hazard problem kicks in: since efforts are not contractible, both would shirk after being matched.

Assume that $\frac{1}{2}\pi_{ii}(H, H) - c(H) > \frac{1}{2}\pi_{ii}(L, H) - c(L)$. This implies that if two agents of the same type are matched, (H, H) is a Nash equilibrium between them. Moreover, the criterion of external stability implies that (H, H) is the only action profile that two agents of the same type will play against each other if they are matched in a stable outcome. This is because even if both choosing L constitutes a Nash equilibrium between them, endogenous matching will make

them form a blocking pair to choose H instead because they both strictly benefit from doing so ($\frac{1}{2}\pi_{ii}(H, H) - c(H) > \frac{1}{2}\pi_{ii}(L, H) - c(L)$ implies $\frac{1}{2}\pi_{ii}(H, H) - c(H) > \frac{1}{2}\pi_{ii}(L, L) - c(L)$).

Assume that $\frac{1}{2}\pi_{ii}(H, H) - c(H) > \frac{1}{2}\pi_{12}(L, L) - c(L)$. This assumption implies that PAM is the unique stable matching.

If we further assume $\pi_{11}(H, H) > \pi_{22}(H, H)$, the globally asymptotic state with respect to $X \setminus \{x | x_i = 0\}$ is $x_1 = 0$, which is inefficient. ■

This example shows that moral hazard under the equal sharing norm can generate inefficiency already at the matching stage. Anticipating the moral hazard problem that arises in cross-type matches, agents prefer to match with their own type, even when cross-type matches are potentially more productive. This matching distortion then translates into long-run evolutionary inefficiency: the population evolves toward homogeneity rather than sustaining a heterogeneous composition that would support more productive cross-type partnerships.

The insights from Example 6 extend to the following general result.

Proposition 6 Suppose there exist types $(i, j) \in N^2$ and a strategy pair $(\sigma_i, \sigma_j) \in \Sigma^2$ such that $(i, j, \sigma_i, \sigma_j) = \operatorname{argmax}_{(k, l, \sigma, \sigma') \in N^2 \times \Sigma^2} \pi_{kl}(\sigma, \sigma')$. **(a)** If (σ_i, σ_j) is a Nash equilibrium of G_{ij} , then the globally asymptotically stable state under (RDE) with respect to X_{++} must be the efficient state. **(b)** If (σ_i, σ_j) is not a Nash equilibrium of G_{ij} , then the globally asymptotically stable state under (RDE) with respect to X_{++} must be inefficient.

Proposition 6 shows that, under the equal sharing norm, evolution can still achieve efficiency when the most productive pair is not vulnerable to moral hazard. Otherwise, moral-hazard concerns inevitably distort efficiency.

5.2 The Market Sharing Norm

Next, we consider the market sharing norm. The definition of a stable outcome is modified as follows to incorporate post-match interactions.

Definition 9 A *stable outcome* is described by a *stable matching* $\mu^*(x)$, a *stable strategy profile* $\gamma^*(x)$, and a vector of *stable payoffs* $w^*(x) = (w_1^*(x), w_2^*(x), \dots, w_n^*(x)) \in \mathbb{R}^n$, such that

- (i) (**individual rationality**) Each person gets at least as much as staying unmatched: For any $i \in N$, $w_i^*(x) \geq 0$;
- (ii) (**surplus efficiency**) Each pair exactly divides the surplus: For any $i \in N$ and $j \in N$ such that $(\sigma_i, \sigma_j) \in \operatorname{supp}\{\gamma_{ij}^*(x)\}$ and $\mu_{ij}^*(x) > 0$, $w_i^*(x) + w_j^*(x) = \pi_{ij}(\sigma_i, \sigma_j) - c_i(\sigma_i) - c_j(\sigma_j)$; and

(iii) (**no blocking pair**) No pair can achieve a sum that is strictly greater than the sum of their current payoffs: For any $i \in N$ and $j \in N$ such that $(\sigma_i, \sigma_j) \in \Sigma^2$, $w_i^*(x) + w_j^*(x) \geq \pi_{ij}(\sigma_i, \sigma_j) - c_i(\sigma_i) - c_j(\sigma_j)$.

Observe that Condition (iii) of Definition 9 immediately implies that if $\mu_{ij}^*(x) > 0$ and $(\sigma_i, \sigma_j) \in \text{supp}\{\gamma_{ij}^*(x)\}$, then $(\sigma_i, \sigma_j) \in \arg \max_{(\sigma, \sigma') \in \Sigma^2} [\pi_{ij}(\sigma, \sigma') - c_i(\sigma) - c_j(\sigma')]$. Critically, this implies that market opportunities effectively eliminate the possibility of moral hazard.

Let $SO^I(x)$ denote the set of stable outcomes in a market with post-match interactions, and let $SO^N(x)$ denote the set of stable outcomes in a market without post-match interactions. We have the following immediate result:

Proposition 7 *Given x , $SO^I(x)$ is identical to $SO^N(x)$ with $f(i, j) \equiv \max_{(\sigma'_i, \sigma'_j) \in \Sigma^2} [\pi_{ij}(\sigma'_i, \sigma'_j) - c_i(\sigma'_i) - c_j(\sigma'_j)]$ for any $i \in N$ and $j \in N$.*

Importantly, Proposition 7 demonstrates that Proposition 1 directly applies, implying that the replicator dynamic (RDI) must lead to efficiency in the long run. Let us use the following example to illustrate Proposition 7.

Example 7 (Evolutionarily stable state under the market sharing norm with two traits)

We keep all the assumptions in Example 6 regarding the payoff structure, but replace the equal sharing norm with the market sharing norm. Without loss of generality, assume $x_1 > x_2$.

Note that $\frac{1}{2}\pi_{ii}(H, H) - c(H) > \frac{1}{2}\pi_{ii}(L, H) - c(L)$ implies $\pi_{ii}(H, H) - 2c(H) > \pi_{ii}(L, L) - 2c(L)$ and $\pi_{ii}(H, H) - 2c(H) > \pi_{ii}(L, H) - c(L) - c(H)$. These together imply that $w_1^*(x) = \frac{1}{2}\pi_{ii}(H, H) - c(H)$ because there always exist matches between two type-1 agents; they must receive the same wage, and condition (iii) in Definition 9 ensures that they both choose H .

If there are matches between the type-1 agents and the type-2 agents, then the payoff of any type-2 agent in such matches equals $w_2^*(x)[\sigma_1, \sigma_2] = \pi_{12}(\sigma_1, \sigma_2) - c(\sigma_1) - c(\sigma_2) - w_1^*(x)$, which depends on both σ_1 and σ_2 . Recall that we have assumed that $\pi_{12}(H, H) - 2c(H)$ is the unique highest joint payoff. This, together with conditions (ii) and (iii) in Definition 9, ensures that (H, H) is the action profile chosen in any cross-type match in a stable outcome. This further implies that $w_2^*(x)[H, H] = \pi_{12}(H, H) - \frac{1}{2}\pi_{11}(H, H) - c(H)$.

The only potential blocking comes from a pair of type-2 agents. However, this cannot happen because $2w_2^*(x)[H, H] \geq \pi_{22}(\sigma'_i, \sigma'_j) - c(\sigma'_i) - c(\sigma'_j)$ for any $(\sigma'_i, \sigma'_j) \in \Delta\{H, L\}$. To see this, first recall that the assumption $\frac{1}{2}\pi_{ii}(H, H) - c(H) > \frac{1}{2}\pi_{ii}(L, H) - c(L)$ implies that $(H, H) = \arg \max_{(e, e')} \pi_{22}(e, e') - c(e) - c(e')$. Second, $2w_2^*(x)[H, H] \geq \pi_{22}(H, H) - 2c(H)$ is equivalent to $2\pi_{12}(H, H) > \pi_{11}(H, H) + \pi_{22}(H, H)$, which holds automatically because we have assumed that $\pi_{12}(H, H) - 2c(H)$ is the unique highest joint payoff.

Hence, the unique stable matching satisfies $\mu_{11}^*(x) = (x_1 - x_2)/2$ and $\mu_{12}^*(x) = x_2$. It is straightforward to see that $w_2^*(x)[H, H] > w_1^*$, implying that the dynamic converges to $x_1 = x_2 = 1/2$. Symmetrically, the analysis for the $x_2 > x_1$ case is similar. We can conclude that the efficient state $x_1 = x_2 = 1/2$ with $\mu_{12}^*(x) = 1/2$ is globally asymptotically stable with respect to X_{++} . ■

Example 7 thus shows that the market sharing norm, through transfers, eliminates the moral hazard problem and leads to short-run efficiency first through matching, then to long-run efficiency through evolution.

5.3 The Evolution of Sharing Norms

Using the results above, we can discuss institutional selection under moral hazard.

Proposition 8 *Suppose there are two communities initially: the market sharing M and the equal sharing E . Suppose there exist types $(i, j) \in N^2$ and a strategy pair $(\sigma_i, \sigma_j) \in \Sigma^2$ such that $(i, j, \sigma_i, \sigma_j) = \operatorname{argmax}_{(k, l, \sigma, \sigma') \in N^2 \times \Sigma^2} \pi_{kl}(\sigma, \sigma')$.*

(i) *If (σ_i, σ_j) is a Nash equilibrium of G_{ij} , then*

$$\lim_{t \rightarrow \infty} q_M(t) > 0, \quad \lim_{t \rightarrow \infty} q_E(t) > 0.$$

(ii) *If (σ_i, σ_j) is not a Nash equilibrium of G_{ij} , then*

$$\lim_{t \rightarrow \infty} q_M(t) > 0, \quad \lim_{t \rightarrow \infty} q_E(t) = 0.$$

Proposition 8 states that (i) when the most efficient pair is not subject to moral hazard, equal sharing and market sharing norms can coexist; (ii) when moral hazard compromises the most efficient pair, however, only the market sharing norm prevails. Propositions 5, 6, and 7 directly imply this proposition.

6 Conclusion

This paper offers an evolutionary perspective on the resilience of equal sharing norms and challenges the assumption that market-based approaches are inherently superior. Through a multi-tier evolutionary framework, we demonstrate how equal sharing, despite its rigidity, supports the emergence of productive traits that facilitate efficient partnerships and thereby enhance long-term productivity. Our findings suggest that the equal sharing norm continues to thrive alongside market mechanisms due to its capacity to shape stable and productive populations. We also show that a

slight deviation from equal sharing, namely an unequal sharing norm close to equal sharing, may persist but only when efficiency requires the population to be homogeneous in productive traits.

Beyond characterizing the evolution of productive traits under a fixed sharing norm, we also allow the sharing norms themselves to evolve across communities. We show that market sharing and equal sharing are both evolutionarily resilient. Although market sharing enjoys a transitional advantage because it achieves efficient matching at each population state, this advantage is finite, and both norms survive in the long run once their communities converge to the same efficient performance. This result provides a direct evolutionary rationale for the persistence of equal sharing alongside market institutions.

Our moral-hazard extension delineates the limits of this resilience. When the most productive partnership is not vulnerable to moral hazard, equal sharing continues to support efficient trait evolution and can coexist with market sharing. When equal division instead undermines effort incentives in the most productive match, equal sharing generates inefficient matching and trait evolution, whereas market sharing can restore incentives through transfers. Under institutional selection, equal sharing then disappears and only market sharing survives. Thus, the evolutionary resilience of equal sharing is substantial but conditional: it depends on whether equal division preserves incentives in the partnerships that matter most for efficiency.

The current model focuses solely on pairwise matching, resulting in a stable distribution that is either monomorphic or bimorphic. A natural extension is to consider team formation. We anticipate that our results will generalize to any fixed team size and stable distributions can exhibit richer trait diversity. It is also an intriguing direction to explore general coalition formations.

A Omitted Proofs

Proof of Lemma 1. The set of feasible one-sided matchings is

$$\mathcal{M}_1(x) = \left\{ \mu = (\mu_{ij}) \left| 2\mu_{ii} + \sum_{j \neq i} \mu_{ij} \leq x_i \forall i \in N \text{ and } \mu_{ij} = \mu_{ji} \forall i, j \in N \right. \right\}.$$

We define an outcome (μ, w) as the specification of a feasible roommate matching μ and an associated vector of payoffs $(w_i)_{i \in N}$, with w_i denoting the payoff of each individual of trait i . These payoffs have to be feasible: The sum of payoffs across the population has to be equal to the total output under the matching μ . In a one-sided matching μ , the total surplus created is

$$\overline{w}^*(\mu) = \sum_{i \in N} \mu_{ii} f(i, i) + \sum_{i \in N} \left[\sum_{j \in N \setminus \{i\}} \frac{\mu_{ij}}{2} f(i, j) \right] = \sum_{i \in N} \left[\mu_{ii} f(i, i) + \sum_{j \in N \setminus \{i\}} \frac{\mu_{ij}}{2} f(i, j) \right].$$

The maximum of the aggregate surplus over the set of feasible one-sided matchings $\mathcal{M}_1(x)$ is

$$P_1(x) = \max_{\mu \in \mathcal{M}_1(x)} \bar{w}^*(\mu). \quad (P_1)$$

Explicitly, this *one-sided primal program* is

$$\begin{aligned} & P_1(x) = \max_{\mu} \bar{w}^*(\mu) \\ \text{subject to} \quad & 2\mu_{ii} + \sum_{j \in N \setminus \{i\}} \mu_{ij} \leq x_i \quad \forall i \in N, \quad (\text{feasibility}) \\ & \mu_{ij} = \mu_{ji} \quad \forall i, j \in N. \quad (\text{symmetry}) \end{aligned}$$

Define the following *one-sided dual program*:

$$\begin{aligned} & D_1(x) = \min_{w=(w_i)_{i \in N}, A=(A_{ij})_{i,j \in N}} \sum_{i \in N} x_i w_i \quad (D_1) \\ \text{subject to} \quad & w_i \geq 0 \quad \forall i \in N \quad (\text{individual rationality}) \\ & w_i + w_j \geq f(i, j) + A_{ij} \quad \forall i, j \in N, \quad (\text{no blocking pair}) \\ & A_{ij} = -A_{ji} \quad \forall i, j \in N. \quad (\text{anti-symmetry}) \end{aligned}$$

Consider an associated two-sided matching problem, which will be important for our subsequent proofs to leverage the well-established tools in the two-sided matching literature. Denote the set of feasible two-sided matchings by

$$\mathcal{M}_2(x, x) = \left\{ \nu = (\nu_{ij}) \left| \sum_{j \in N} \nu_{ij} \leq x_i \forall i \in N \text{ and } \sum_{i \in N} \nu_{ij} \leq x_j \forall j \in N \right. \right\},$$

and the total surplus of a two-sided matching ν by

$$\bar{w}_2^*(\nu) = \sum_{i,j \in N} \nu_{ij} f(i, j).$$

A stable outcome in the two-sided matching setting is defined analogously: (ν, u, v) is a *two-sided stable outcome* that consists of a feasible two-sided matching $\nu = (\nu_{ij})_{i,j \in N}$ and two-sided payoff vectors $u = (u_i)_{i \in N}$ and $v = (v_j)_{j \in N}$ if (i) (individual rationality) $u_i \geq 0$ for any $i \in N$ and $v_j \geq 0$ for any $j \in N$, (ii) (no blocking pair) $u_i + v_j \geq f(i, j)$ for any $i, j \in N$, and (iii) (surplus efficiency) $u_i + v_j = f(i, j)$ if $\nu_{ij} > 0$.

The two-sided surplus maximization problem of population state $(x/2, x/2)$ —the *two-sided*

primal program—associated with the one-sided primal program (P_1) of population state x is

$$P_2(x/2, x/2) = \max_{\nu \in \mathcal{M}_2(x/2, x/2)} \bar{w}_2^*(\nu). \quad (P_2)$$

Explicitly,

$$\begin{aligned} & P_2(x/2, x/2) = \max_{\nu} \bar{w}_2^*(\nu) \\ \text{subject to} \quad & \sum_{j \in N} \nu_{ij} \leq x_i \forall i \in N \text{ and } \sum_{i \in N} \nu_{ij} \leq x_j \forall j \in N. \end{aligned} \quad (\text{feasibility})$$

Note that the symmetry constraints in the one-sided primal program are absent in the two-sided primal program. The **two-sided dual program** is

$$\begin{aligned} D_2(x/2, x/2) &= \min_{u, v} \sum_{i \in N} \frac{1}{2} x_i u_i + \sum_{j \in N} \frac{1}{2} x_j v_j \quad (D_2) \\ \text{subject to} \quad & u_i \geq 0 \forall i \in N \text{ and } v_j \geq 0 \forall j \in N, \quad (\text{individual rationality}) \\ & u_i + v_j \geq f(i, j) \quad \forall i, j \in N. \quad (\text{no blocking pair}) \end{aligned}$$

By [Shapley and Shubik \(1972\)](#) and [Gretsky, Ostroy, and Zame \(1992\)](#), there exist two-sided stable matchings ν and they coincide with the solution of the two-sided primal program (P_2). Moreover, the associated payoffs (u, v) solve the two-sided dual program (D_2). Finally, for any stable matching ν , $\nu_{ij} > 0$ implies $u_i + v_j = f(i, j)$, $\nu_{i0} > 0$ implies $u_i = 0$, and ν_{0j} implies $v_j = 0$.

Note that the two-sided matching problem is a particular case of the one-sided matching problem: If $f(i, j) = -\infty$ whenever i and j belong to the same side, then any optimal or stable matching will be bipartite.

Claim: The values of the one-sided primal program with population state $x \in X_{++}$ and the associated two-sided primal program with population state $(x/2, x/2)$ coincide:

$$P_1(x) = P_2(x/2, x/2).$$

Consider a matching $\mu \in \mathcal{M}_1(x)$. Define ν such that $\nu_{ij} = \mu_{ij}/2$ if $i \neq j$ and $\nu_{ii} = \mu_{ii}$. For any $i \in N$,

$$\sum_{j \in N} \nu_{ij} = \nu_{ii} + \sum_{j \neq i} \nu_{ij} = \mu_{ii} + \sum_{j \neq i} \mu_{ij}/2 \leq x_i/2,$$

and

$$\sum_{j \in N} \nu_{ji} = \nu_{ii} + \sum_{j \neq i} \nu_{ji} = \mu_{ii} + \sum_{j \neq i} \mu_{ji}/2 \leq x_i/2,$$

so $\nu \in \mathcal{M}_2(x/2, x/2)$, as the feasibility constraint for $\nu \in \mathcal{M}_2(x/2, x/2)$ is satisfied.

$$\bar{w}^*(\mu) = \sum_{i \in N} \mu_{ii} f(i, i) + \sum_{i \in N} \left[\sum_{j \in N \setminus \{i\}} \frac{\mu_{ij}}{2} f(i, j) \right] = \sum_{i \in N} \left[\sum_{j \in N} \nu_{ij} f(i, j) \right] = \bar{w}_2^*(\nu).$$

Hence, ν , a feasible two-sided matching in $(x/2, x/2)$, achieves the maximum surplus of one-sided matching under population state x (achieved by the optimal solution μ). We have

$$P_1(x) \leq P_2(x/2, x/2). \quad (1)$$

Conversely, let (ν_{ij}) maximize aggregate surplus over $\mathcal{M}_2(x/2, x/2)$. By the symmetry of f , (ν_{ji}) is also a maximizer. Since (P_2) is a linear program, $\nu'_{ij} = (\nu_{ij} + \nu_{ji})/2$ also maximizes it. Define μ such that $\mu'_{ij} = 2\nu'_{ij}$ if $i \neq j$ and $\mu'_{ii} = \nu'_{ii}$. Then for any $i \in N$,

$$2\mu'_{ii} + \sum_{j \neq i} \mu'_{ij} = 2\nu_{ii} + \sum_{j \neq i} (\nu_{ij} + \nu_{ji}) = \sum_{j \in N} \nu_{ij} + \sum_{j \in N} \nu_{ji} \leq x_i,$$

where $\sum_{j \in N} \nu_{ij} \leq x_i/2$ and $\sum_{j \in N} \nu_{ji} \leq x_i/2$ follow from the feasibility constraints of $\nu \in \mathcal{M}_2(x/2, x/2)$. Hence, $\mu' \in \mathcal{M}_1(x)$. The aggregate surplus under one-sided matching μ' is

$$\sum_{i \in N} \mu'_{ii} f(i, i) + \sum_{i \in N} \left[\sum_{j \neq i} \frac{\mu'_{ij}}{2} f(i, j) \right] = \sum_{i \in N} \nu'_{ii} f(i, i) + \sum_{i \in N} \left[\sum_{j \neq i} \nu'_{ij} f(i, j) \right] = \sum_{i \in N} \sum_{j \in N} \nu'_{ij} f(i, j).$$

Hence, μ' , a feasible one-sided matching in population state x , achieves the maximal two-sided surplus in population state $(x/2, x/2)$ (achieved by the optimal solution ν'). Hence,

$$P_1(x) \geq P_2(x/2, x/2). \quad (2)$$

Combining (1) and (2), we have $P_1(x) = P_2(x/2, x/2)$.

By the claim, the values of one-sided primal program (P_1) $P_1(x)$ and two-sided primal program (P_2) $P_2(x/2, x/2)$ coincide. Let $\nu = (\nu_{ij})$ denote a solution of the two-sided primal program. By the duality result of [Shapley and Shubik \(1972\)](#) and [Gretsky, Ostroy, and Zame \(1992\)](#), there exist two-sided stable payoff vectors (u_i) and (v_j) that solve the two-sided dual program (D_2) $D_2(x/2, x/2)$, namely, such that $u_i \geq 0$ for any $i \in N$, $v_j \geq 0$ for any $j \in N$, $u_i + v_j \geq f(i, j)$ for any $i, j \in N$, and $u_i + v_j = f(i, j)$ whenever $\nu_{ij} > 0$.

Define $\mu_{ij} = \nu_{ij} + \nu_{ji}$ when $i \neq j$, $\mu_{ii} = \nu_{ii}$, and $w_i = (u_i + v_i)/2$ for any $i \in N$. We have

$w_i \geq 0$, $w_i + w_j \geq f(i, j)$, $w_i + w_j = f(i, j)$ whenever $\mu_{ij} > 0$, and

$$\sum_{i \in N} x_i w_i = \frac{1}{2} \left[\sum_{i \in N} x_i u_i + \sum_{i \in N} x_i v_i \right] = P_2(x/2, x/2) = P_1(x) = \bar{w}^*(\mu).$$

Hence, (μ, w) is a one-sided stable outcome. ■

Proof of Lemma 2. First, fix a population state $x \in X_{++}$. The set of stable one-sided payoffs, $W^*(x)$, is convex. Fix two stable payoff vectors $w^\spadesuit \in W^*(x)$ and $w^\diamond \in W^*(x)$. Take any $\lambda \in (0, 1)$. We have (i) $\lambda w_i^\spadesuit + (1 - \lambda)w_i^\diamond \geq 0$ for any $i \in N$, because $w_i^\spadesuit \geq 0$ and $w_i^\diamond \geq 0$ for any $i \in N$, (ii) $[\lambda w_i^\spadesuit + (1 - \lambda)w_i^\diamond] + [\lambda w_j^\spadesuit + (1 - \lambda)w_j^\diamond] \geq f(i, j)$ for any $i, j \in N$, because $w_i^\spadesuit + w_j^\spadesuit \geq f(i, j)$ and $w_i^\diamond + w_j^\diamond \geq f(i, j) \geq f(i, j)$.

Second, fix a population state $x \in X_{++}$. The set of stable one-sided payoffs, $W^*(x)$, is compact. Since $W^*(x)$ is a set in $\mathbb{R}^{|I|}$, it suffices to show that $W^*(x)$ is closed and bounded. Each element of $W^*(x)$ is bounded due to the surplus function being bounded. The set is also closed, because all individual rationality and no-blocking-pair constraints are non-strict inequalities.

Third, the correspondence, $W : X_{++} \rightrightarrows \mathbb{R}^{|I|}$, that specifies the set of stable one-sided payoffs $W^*(x)$ for population state $x \in X_{++}$, is upper-hemicontinuous. Proof: $W : X_{++} \rightrightarrows \mathbb{R}^{|I|}$ is upper-hemicontinuous if and only if $x^k \rightarrow x$, $w^k \in W(x^k)$, and $w^k \rightarrow w$ imply $w \in W^*(x)$. It suffices to show that $\bar{w}^*(x) = \sum_{i \in N} w_i x_i$ and that w satisfies $w_i \geq 0$ for any $i \in N$ and $w_i + w_j \geq f(i, j)$ for any $i, j \in N$. First, we show that $\bar{w}^*(x) = \sum_{i \in N} w_i x_i$.

$$\begin{aligned} & \left| \sum_{i \in N} w_i x_i - \bar{w}^*(x) \right| \\ &= \left| \sum_{i \in N} w_i x_i - \sum_{i \in N} w_i^k x_i^k + \bar{w}^*(x^k) - \bar{w}^*(x) \right| \\ &\leq \left| \sum_{i \in N} w_i x_i - \sum_{i \in N} w_i^k x_i^k \right| + |\bar{w}^*(x^k) - \bar{w}^*(x)| \\ &\leq \left| \sum_{i \in N} w_i x_i - \sum_{i \in N} w_i x_i^k \right| + \left| \sum_{i \in N} w_i x_i^k - \sum_{i \in N} w_i^k x_i^k \right| + |\bar{w}^*(x^k) - \bar{w}^*(x)| \\ &\leq \left| \sum_{i \in N} w_i x_i - \sum_{i \in N} w_i x_i^k \right| + \|w - w^k\| + |\bar{w}^*(x^k) - \bar{w}^*(x)|. \end{aligned}$$

Fix ϵ . Since $x^k \rightarrow x$, there is a $k_1(\epsilon)$ such that all $k > k_1(\epsilon)$, $|\sum_{i \in N} w_i x_i - \sum_{i \in N} w_i x_i^k| \leq \epsilon/3$. Since $w^k \rightarrow w$, there is a $k_2(\epsilon)$ such that for all $k > k_2(\epsilon)$, $\|w - w^k\| \leq \epsilon/3$. Since $x^k \rightarrow x$, $|\bar{w}^*(x^k) - \bar{w}^*(x)| \leq \epsilon/3$. Therefore, $|\sum_{i \in N} w_i x_i - \bar{w}^*(x)| \leq \epsilon$ for any $\epsilon > 0$.

We also need to show that w satisfies the stability conditions. Suppose otherwise: $w_i + w_j < f(i, j)$ for a positive measure of (i, j) or $w_i < 0$ for some $i \in N$. Then $\lim_{k \rightarrow \infty} \|w - w^k\| > 0$, a contradiction to the assumption that $w^k \rightarrow w$. $\blacksquare$

In what follows, we provide the theorems we will depend on to use the Lyapunov method to show convergence and stability of the dynamics.

Theorem 2 (*Sandholm, 2010, Theorem 7.B.2*) *Let $A \subseteq X$ be closed and let $Y \subseteq X$ be a neighborhood of A . Let $L : Y \rightarrow \mathbb{R}_+$ be Lipschitz continuous with $L^{-1}(0) = A$. If each solution $\{x^t\}_{t \geq 0}$ of $\dot{x} \in V^{RDI}(x)$ (or $\dot{x} = V^{RDE}(x)$) satisfies $\dot{L}(x^t) \leq 0$ for almost all $t \geq 0$, then A is Lyapunov stable.*

Theorem 2 states that the existence of a Lipschitz continuous Lyapunov function $L(x)$ guarantees that $L^{-1}(0)$, the set of population states that minimize $L(x)$, is Lyapunov stable.

Theorem 3 (*Sandholm, 2010, Theorem 7.B.4*) *Let $Y \subset X$ be relatively open and inescapable under (RDI). Let $L : Y \rightarrow \mathbb{R}$ be C^1 and satisfy (i) $\frac{\partial L}{\partial v}(x) \equiv \nabla L(x)'v \leq 0$ for all $v \in V^{RDI}(x)$ and $x \in Y$ and (ii) $[\mathbf{0} \notin V^{RDI}(x) \implies \frac{\partial L}{\partial v}(x) < 0]$ for all $v \in V^{RDI}(x)$ and $x \in Y$. Then, for all solutions $\{x^t\}$ of (RDI) with $x_0 \in Y$, $\omega(\{x^t\}) \subseteq \{x \in Y : \mathbf{0} \in V(x)\}$.*

Theorem 3 shows that the existence of a C^1 Lyapunov function $L(x)$ guarantees the solution trajectories converge to the states that minimize $L(x)$. Combining Theorems 2 and 3 gives us the globally asymptotic stability of the set of population states that minimize $L(x)$ under (RDI).

Proof of Proposition 1. In the first case, $L^{RDI}(x) = -\log x_i$ is a C^1 function. Moreover, for any x such that $x_i > 0$, and for any $v = x \cdot [w^*(x) - \bar{w}^*(x)]$, we have

$$\frac{\partial L^{RDI}}{\partial v}(x) = \nabla L^{RDI}(x)'v = -\left(\frac{1}{x_i}\right) x_i [w_i^*(x) - \bar{w}^*(x)] = -[w_i^*(x) - \bar{w}^*(x)].$$

By the definition of stable matching, $w_i^*(x) \geq f(i, i)/2$. By the fact that $\operatorname{argmax}_{(k, l)} f(k, l) = \{(i, i)\}$, $f(i, i)/2 > \bar{w}^*(x)$ as long as $x \neq e_i$. Therefore, we have $\frac{\partial L^{RDI}}{\partial v}(x) < 0$. At $x^* = e_i$, the stable matching is given by $\mu_{ii}(e_i) = 0.5$, where $w_i^*(e_i) = \bar{w}^*(e_i) = f(1, 1)/2$. Hence, $V^{RDI}(e^1) = \{\mathbf{0}\}$. Also, $L^{-1}(0) = \{e_i\}$. By Theorems 2 and 3, e_i is the globally asymptotically stable state with respect to $X \setminus \{x | x_i = 0\}$. In addition, $\operatorname{argmax}_{(k, l)} f(k, l) = \{(i, i)\}$ implies that e_i is the unique efficient state.

In the second case, $L^{RDI}(x) = -[\log(2x_i) + \log(2x_j)]$ is a C^1 function. Moreover, for any x such that $x_i > 0$ and $x_j > 0$, and for any $v = x \cdot [w^*(x) - \bar{w}^*(x)]$, we have

$$\frac{\partial L^{RDI}}{\partial v}(x) = \nabla L^{RDI}(x)'v$$

$$\begin{aligned}
&= -\left(\frac{1}{x_i}\right) x_i [w_i^*(x) - \bar{w}^*(x)] - \left(\frac{1}{x_j}\right) x_j [w_j^*(x) - \bar{w}^*(x)] \\
&= -[(w_i^*(x) + w_j^*(x)) - 2\bar{w}^*(x)].
\end{aligned}$$

By the definition of stable matching, $w_i^*(x) + w_j^*(x) \geq f(i, j)$. By $\operatorname{argmax}_{(k,l)} f(k, l) = \{(i, j)\}$, we have $f(i, j)/2 > \bar{w}^*(x)$. Therefore, we have $\frac{\partial L^{RDI}}{\partial v}(x) < 0$. At x^* such that $x_i^* = x_j^* = 0.5$, the stable matching is given by $\mu_{ij}(x^*) = 0.5$, where $w^*(x^*)$ that satisfies $w_i^*(x^*) = w_j^*(x^*) = f(i, j)/2$ is a possible stable payoff vector. Hence $\mathbf{0} \in V^{RDI}(x^*)$. Also, $L^{-1}(0) = \{x^*\}$. By Theorems 2 and 3, x^* is the globally asymptotically stable state with respect to $X \setminus \{x | x_i = 0 \text{ or } x_j = 0 \text{ or both}\}$. In addition, $\operatorname{argmax}_{(k,l)} f(k, l) = \{(i, j)\}$ implies that x^* is the unique efficient state. ■

Theorem 4 (*Sandholm, 2010, Corollary 7.B.6*) *Let $A \subseteq X$ be closed and let $Y \subseteq X$ be a neighborhood of A . Let $L : Y \rightarrow \mathbb{R}$ be C^1 with $L^{-1}(0) = A$. If $\dot{L}(x) < 0$ for all $x \in Y - A$, then A is asymptotically stable under $\dot{x} = V^{RDE}(x)$. If in addition, $Y = X$, then A is globally asymptotically stable under $\dot{x} = V^{RDE}(x)$.*

Theorem 4 gives us globally asymptotic stability of the set of population states that minimize $L(x)$ under (RDE).

Proof of Proposition 2. In the first case, $\operatorname{argmax}_{(k,l)} f(k, l) = \{(1, 1)\}$. implies that $f(1, 1) > f(1, i)$ for any $i \neq 1$. Hence, as long as $x_1 > 0$, all trait-1 agents must match among themselves and $u_1^*(x) = u_{11} = f(1, 1)/2$ is the highest payoff an agent can obtain in the population.

We want to show that $L^{RDE}(x) = -\log x_1$ serves as a Lyapunov function for the replicator dynamic. First, it is a C^1 function with $L^{RDE}(x^*) = 0$. Furthermore, for any x^t such that $x_1^t > 0$,

$$\begin{aligned}
dL^{RDE}(x^t)/dt &= \nabla L^{RDE}(x^t)' \dot{x}^t \\
&= -\left(\frac{1}{x_1^t}\right) x_1^t [u_1^*(x^t) - \bar{u}^*(x^t)] \\
&= \bar{u}^*(x^t) - u_1^*(x^t) \\
&< 0 \text{ as long as } x^t \neq e_1.
\end{aligned}$$

Hence, according to Theorem 4, x^* is globally asymptotically stable with respect to $X \setminus \{x | x_1 = 0\}$.

In the second case, $\operatorname{argmax}_{(k,l)} f(k, l) = \{(i, j)\}$ implies that $f(i, j) > f(i, i)$ and $f(i, j) > f(j, j)$. Hence, these two traits of agents will match with each other to the greatest extent possible. Without loss of generality, assume that $0 < x_i^0 < x_j^0$. Then all trait- i agents are matched with trait- j agents, while the remaining trait- j agents match otherwise. In this case, $u_i^*(x^0) = u_{ij} = f(i, j)/2$, which is the largest payoff an agent can get.

Let $r(t) = x_i^t/x_j^t$. Observe that under the replicator dynamic, $\frac{dr(t)}{dt} = \frac{x_i^t}{x_j^t} [u_i^*(x^t) - u_j^*(x^t)] > 0$ as long as $x_i^t < x_j^t$ so that $u_i^*(x^t) - u_j^*(x^t) > 0$. Also, once $x_i^t = x_j^t$, $r(t) = 1$ and $u_i^*(x^t) = u_j^*(x^t) = f(i, j)/2$ because all trait- i agents are exactly matched with all trait- j agents. Hence, $\frac{dr(t)}{dt} = 0$. In sum, $r(t)$ is a strictly increasing function bounded by 1, implying that $\lim_{t \rightarrow \infty} r(t) = 1$. Define $Y = \{x | x_i = x_j > 0\}$. This part of the proof shows that the dynamic always converges to Y from $X \setminus \{x | x_i = 0 \text{ or } x_j = 0 \text{ or both}\}$.

Next, we want to show that x^* is asymptotically stable with respect to Y . $L^{RDE}(x) = -[\log(2x_j) + \log(2x_i)]$ is a C_1 function in Y with $L^{RDE}(x^*) = 0$. In addition,

$$\begin{aligned} dL^{RDE}(x^t)/dt &= \nabla L^{RDE}(x^t)' \dot{x}^t \\ &= -\left(\frac{1}{2x_i^t}\right) x_i^t [u_i^*(x^t) - \bar{u}^*(x^t)] - \left(\frac{1}{2x_j^t}\right) x_j^t [u_j^*(x^t) - \bar{u}^*(x^t)] \\ &= \bar{u}^*(x^t) - [u_i^*(x^t) + u_j^*(x^t)] / 2 \\ &= \bar{u}^*(x^t) - f(i, j)/2 \\ &< 0 \text{ as long as } x^t \neq x^*. \end{aligned}$$

Hence, $L^{RDE}(x)$ is a Lyapunov function for the replicator dynamic on Y and once the dynamic reaches Y , the dynamic converges to x^* according to Theorem 4. In sum, we can conclude that x^* is globally asymptotically stable with respect to $X \setminus \{x | x_i = 0 \text{ or } x_j = 0 \text{ or both}\}$. ■

Proof of Proposition 3. We will prove by construction. First, let $f(i, i) = c$ for some constant $c > 0$, for any $i \neq 1$. For any $1 < i < j$, let $f(i, j) = c + \nu_{ij}$, where $0 < \nu_{ij} < c[1/(2\alpha_{ji}) - 1]$. We must have that for any $1 < i < j$,

$$u_{ji} = \alpha_{ji}(c + \nu_{ij}) < u_{jj} = u_{ii} = c/2 < u_{ij} = \alpha_{ij}(c + \nu_{ij}).$$

This implies that in any stable matching, no $i - j$ match will be formed.

Next, let $f(1, i) = c + \epsilon_i$ for any $i \neq 1$, where $0 < \epsilon_i < c[1/(2\alpha_{i1}) - 1]$. Let $\epsilon = \min_i \epsilon_i$ and let $f(1, 1) = c + \epsilon/2$. Then we have, for any $i \neq 1$,

$$u_{i1} = \alpha_{i1}(c + \epsilon_i) < c/2 = u_{ii} < u_{11} = (c + \epsilon/2)/2 < (c + \epsilon_i)/2 < \alpha_{1i}(c + \epsilon_i) = u_{1i}.$$

This implies that in any stable matching, no $1 - i$ match will be formed. Hence, the unique stable matching must be PAM and e_1 is the globally asymptotically stable state $X \setminus \{x | x_1 = 0\}$. However, it is not efficient because $f(1, 1) < f(1, i)$ for any $i \neq 1$. ■

Proof of Proposition 4. In the first case, $\operatorname{argmax}_{(k,l)} f(k, l) = \{(i, j)\}$ implies that the efficient state must take the form of x^* with $x_i^* = x_j^* = 0.5$. However, since $\alpha_{ij} \neq \alpha_{ji}$,

$u_i(x^*) = f(i, j)\alpha_{ij} \neq f(i, j)\alpha_{ji} = u_j(x^*)$. Hence, x^* is unstable.

In the second case, $\operatorname{argmax}_{(k,l)} f(k, l) = \{(i, i)\}$ implies that $f(i, i) > f(k, l)$ for any k, l such that not both of them equal i . If $k > l$, since $\alpha_{kl} < 1/2$, we must have $f(i, i)/2 > f(k, l)\alpha_{kl}$. If $k = l$, then $f(i, i)/2 > f(k, k)/2$. If $k < l$, since $f(i, i) > f(k, l)$, there exists $\alpha_{kl} > 1/2$ such that $f(i, i)/2 > f(k, l)\alpha_{kl}$. Hence, given such α_{kl} , as long as $x_i > 0$, all trait- i agents must match among themselves and $u_i^*(x) = u_{ii} = f(i, i)/2$ is the highest payoff an agent can obtain in the population. Then we can repeat the same argument as in Proposition 2 to show that $x^* = e_i$ is globally asymptotically with respect to $X \setminus \{x | x_i = 0\}$. ■

Proof of Proposition 5. The proof is based on the following two claims.

- (a) If $\lim_{t \rightarrow \infty} V_s(x^s(t)) = V_s^*$, $\lim_{t \rightarrow \infty} V_r(x^r(t)) = V_r^*$, and $V_s^* > V_r^*$, then $\frac{q_s(t)}{q_r(t)} \rightarrow \infty$.
- (b) If $V_s^* = V_r^*$, and $\int_0^\infty [V_s(x^s(t)) - V_r(x^r(t))] dt$ is finite, then

$$\lim_{t \rightarrow \infty} \frac{q_s(t)}{q_r(t)} = \frac{q_s(0)}{q_r(0)} \exp \left\{ \varepsilon \int_0^\infty [V_s(x^s(t)) - V_r(x^r(t))] dt \right\}.$$

For any norm $s \in \mathbf{S}$, the norm-level dynamic implies

$$\frac{\dot{q}_s(t)}{q_s(t)} = \varepsilon [V_s(x^s(t)) - \bar{V}(t)],$$

which further implies that for any $s, r \in \mathbf{S}$

$$\frac{d}{dt} \log \frac{q_s(t)}{q_r(t)} = \varepsilon [V_s(x^s(t)) - V_r(x^r(t))].$$

Integrating from 0 to t , we obtain

$$\log \frac{q_s(t)}{q_r(t)} = \log \frac{q_s(0)}{q_r(0)} + \varepsilon \int_0^t [V_s(x^s(\tau)) - V_r(x^r(\tau))] d\tau.$$

Exponentiating both sides yields

$$\frac{q_s(t)}{q_r(t)} = \frac{q_s(0)}{q_r(0)} \exp \left\{ \varepsilon \int_0^t [V_s(x^s(\tau)) - V_r(x^r(\tau))] d\tau \right\}.$$

Now suppose $V_s(x^s(t)) \rightarrow V_s^*$, $V_r(x^r(t)) \rightarrow V_r^*$, and $V_s^* > V_r^*$. Then there exist some $\eta > 0$ and some T such that, for all $t \geq T$, $V_s(x^s(t)) - V_r(x^r(t)) > \eta$. Hence,

$$\int_0^t [V_s(x^s(\tau)) - V_r(x^r(\tau))] d\tau \rightarrow \infty \implies \frac{q_s(t)}{q_r(t)} \rightarrow \infty.$$

Second, suppose that $V_s^* = V_r^*$ and that $\int_0^\infty [V_s(x^s(t)) - V_r(x^r(t))] dt$ is finite. Taking the limit in the relative-share formula gives

$$\lim_{t \rightarrow \infty} \frac{q_s(t)}{q_r(t)} = \frac{q_s(0)}{q_r(0)} \exp \left\{ \varepsilon \int_0^\infty [V_s(x^s(t)) - V_r(x^r(t))] dt \right\}.$$

This completes the claim.

We have established that under both the equal sharing norm and the market sharing norm, type evolution leads to the efficient state, while under the unequal sharing norm it does not necessarily do so. Hence, we must have $V_N^* \leq V_E^* = V_M^*$. This immediately implies that $\lim_{t \rightarrow \infty} q_N(t) \geq 0$.

Next, we show the rest of the proposition. We already know that $V_E^* = V_M^*$. We want to show $\int_0^\infty [V_s(x^M(t)) - V_r(x^E(t))] dt$ is finite because its finiteness ensures that the equal sharing norm is not eliminated. A sufficient condition for this is that the replicator dynamics in communities M and E both converge exponentially to the efficient state.

Case 1: $\operatorname{argmax}_{(k,l)} f(k,l) = \{(i,i)\}$ for some i . In this case, since at the efficient state $x_i = 1$, type i has a strict payoff advantage, for every $j \neq i$, $x_j(t)/x_i(t)$ decays exponentially near $x_i = 1$, implying exponential convergence.

Case 2: $\operatorname{argmax}_{(k,l)} f(k,l) = \{(i,j)\}$ for some i and $j \neq i$. In this case, the efficient state is $x_i = x_j = 1/2$. The dynamics exponentially converges to the efficient state if (i) at $x_i = x_j = 1/2$, types i and j have a strict payoff advantage compared to all other types, which is automatically true, and (ii) the dynamic restricted to the support $\{i,j\}$ converges exponentially or in finite time to the efficient state.

For the market sharing norm, when $x_i^M + x_j^M = 1$, (RDI) takes the form of

$$\dot{x}_i^M = \begin{cases} x_i^M x_j^M (f(i,j) - f(j,j)) & \text{if } 0 < x_i^M < \frac{1}{2}, \\ x_i^M x_j^M (f(i,i) - f(i,j)) & \text{if } \frac{1}{2} < x_i^M < 1. \end{cases}$$

Let $T^{inf} = \inf\{t \geq 0, x_i^M(t) = 1/2\}$ be the first hitting time at which the dynamics reaches the efficient state. When $0 < x_i^M(0) < 1/2$, $T^{inf} = \frac{1}{f(i,j) - f(j,j)} \log \frac{1 - x_i^M(0)}{x_i^M(0)} < \infty$. When $1/2 < x_i^M(0) < 1$, $T^{inf} = \frac{1}{f(i,j) - f(i,i)} \log \frac{x_i^M(0)}{1 - x_i^M(0)} < \infty$.

For the equal sharing norm, when $x_i^E + x_j^E = 1$, (RDE) takes the form of

$$\dot{x}_i^E = \begin{cases} x_i^E (x_j^E - x_i^E) (f(i,j) - f(j,j))/2 & \text{if } 0 < x_i^E \leq \frac{1}{2}, \\ x_j^E (x_i^E - x_j^E) (f(i,i) - f(i,j))/2 & \text{if } \frac{1}{2} < x_i^E < 1. \end{cases}$$

Since $f(i,j) > f(j,j)$ and $f(i,j) > f(i,i)$, the vector field points toward $x_i^E = 1/2$ from both

sides. Moreover, in a neighborhood of $x_i^E = 1/2$, the deviation $|x_i^E(t) - 1/2|$ satisfies

$$\frac{d}{dt} \left| x_i^E(t) - \frac{1}{2} \right| = \begin{cases} -x_i^E(t)[f(i, j) - f(j, j)]|x_i^E(t) - \frac{1}{2}| & \text{if } x_i^E(t) \leq 1/2, \\ -(1 - x_i^E(t))[f(i, j) - f(i, i)]|x_i^E(t) - \frac{1}{2}| & \text{if } x_i^E(t) > 1/2. \end{cases}$$

This implies that

$$\frac{d}{dt} \left| x_i^E(t) - \frac{1}{2} \right| \leq -c \left| x_i^E(t) - \frac{1}{2} \right|$$

for some $c > 0$. Hence, $x_i^E(t)$ converges to $1/2$ exponentially. ■

Proof of Proposition 6. In case (a), condition (iii) in Definition 8 ensures that in any stable outcome, if $\mu_{ij}^*(x) > 0$, then $\gamma_{ij}^*(x)[(\sigma_i, \sigma_j)] = 1$. This further implies that type- i and type- j agents will match with each other to the greatest extent possible. Then, by the proof of Proposition 2, $x_i = x_j = \frac{1}{2}$ is the globally asymptotically stable state under (RDE) with respect to X_{++} (note that i and j can be identical). In case (b), condition (ii) of Definition 8 already rules out the possibility that a positive mass of $i - j$ matches playing (σ_i, σ_j) . Hence, regardless of where (RDE) converges to (even if it is $x_i = x_j = \frac{1}{2}$), the long-run outcome of evolution is inefficient. ■

Proof of Proposition 7. Suppose $(\mu^*(x), \gamma^*(x), w^*(x)) \in SO^I(x)$. Condition (iii) of Definition 9 implies that for any $i \in N$ and $j \in N$, $w_i^*(x) + w_j^*(x) \geq \max_{(\tilde{\sigma}_i, \tilde{\sigma}_j) \in \Sigma^2} \pi_{ij}(\tilde{\sigma}_i, \tilde{\sigma}_j) - c_i(\tilde{\sigma}_i) - c_j(\tilde{\sigma}_j) = f(i, j)$. This further implies that if $\mu_{ij}^*(x) > 0$, $w_i^*(x) + w_j^*(x) = f(i, j)$. In addition, we must have $w_i^*(x) \geq 0$ for any $i \in N$. Hence, $(\mu^*(x), w^*(x))$ satisfies the definition of a stable outcome in Definition 1, implying that it belongs to SO^N .

Suppose $(\mu^*(x), w^*(x)) \in SO^N$ with $f(i, j) = \max_{(\tilde{\sigma}_i, \tilde{\sigma}_j) \in \Sigma^2} \pi_{ij}(\tilde{\sigma}_i, \tilde{\sigma}_j) - c_i(\tilde{\sigma}_i) - c_j(\tilde{\sigma}_j)$. Let $\gamma^*(x)$ satisfy the following criterion. If $\mu_{ij}^*(x) > 0$, then any $(\sigma_i, \sigma_j) \in \text{supp}\{\gamma_{ij}^*(x)\}$ satisfies that $(\sigma_i, \sigma_j) \in \arg \max_{(\tilde{\sigma}_i, \tilde{\sigma}_j) \in \Sigma^2} [\pi_{ij}(\tilde{\sigma}_i, \tilde{\sigma}_j) - c_i(\tilde{\sigma}_i) - c_j(\tilde{\sigma}_j)]$. This implies that if $\mu_{ij}^*(x) > 0$, $w_i^*(x) + w_j^*(x) = f(i, j) = \pi_{ij}(\sigma_i, \sigma_j) - c_i(\sigma_i) - c_j(\sigma_j)$ for any $(\sigma_i, \sigma_j) \in \text{supp}\{\gamma_{ij}^*(x)\}$. This shows that $(\mu^*(x), \gamma^*(x), w^*(x))$ satisfy condition (ii) in Definition 9. Moreover, $w_i^*(x) + w_j^*(x) \geq f(i, j) = \max_{(\tilde{\sigma}_i, \tilde{\sigma}_j) \in \Sigma^2} [\pi_{ij}(\tilde{\sigma}_i, \tilde{\sigma}_j) - c_i(\tilde{\sigma}_i) - c_j(\tilde{\sigma}_j)]$ for any $i, j \in N$ implies that condition (iii) in Definition 9 is satisfied. Finally, we have $w_i^*(x) \geq 0$ for any $i \in N$. Hence, $(\mu^*(x), \gamma^*(x), w^*(x)) \in SO^N$. ■

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